

POLITECNICO
MILANO 1863

Innovative ultra-rapid trans-oceanic delivery



TBD - To Be Delivered

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Abstract

Fools say that they learn by
experience. I prefer to profit by
others' experience

Otto Von Bismarck

The purpose of this report is to present the preliminary analysis of TBD an ICBM, developed with the task of carrying a payload of 100 kg maximum, for a flexible range of 7000-10000 km. An ultra-fast solution is proposed for trans-oceanic deliveries, thus transporting the cargo from the launch site to 3 possible landing sites in a timeframe of 6 hours. It is avoided using propellants and solutions that could cause damage to the environment in the event of accidents. Furthermore, it was arranged that the trajectory followed would allow the division of the stages over uninhabited areas. It was decided to use high TRL components so that the solution could be developed and marketable within 5 years. The proposed missile uses three solid boosters already available on the market and performs the three possible trajectories in a time frame of less than an hour with no reusability for the propulsion units, deploying the payload by parachute re-entry.

Nomenclature

α	Angle of attack	$[rad]$
A	Axial force	$[N]$
A_e	Nozzle exit area	$[m^2]$
A_{ref}	Reference area	$[m^2]$
C_d	Drag coefficient	$[-]$
C_L	Lift coefficient	$[-]$
cg	Center of gravity	$[-]$
Cp	Center of pressure	$[-]$
γ	Flight path angle	$[rad]$
$\dot{\gamma}$	Rate of change of flight path angle	$[rad]$
δ	Steering angle	$[rad]$
d	Diameter	$[m]$
d_F	Fairing diameter	$[m]$
Δv	Change in velocity	$[m/s]$
ε_s	Inert mass fraction	$[-]$
F_{drag}	Drag force	$[N]$
g_0	Gravitational acceleration	$[m/s^2]$
h	Altitude	$[m]$
I_s	Gravimetric specific impulse	$[s]$
I_{tot}	Total impulse	$[Ns]$
l	Length	$[m]$
l/d	Fineness ratio	$[-]$
λ	Lagrangian multiplier	$[-]$
m_{avio}	Avionics mass	$[kg]$
m_{pay}	Payload mass	$[kg]$
m_{pb}	Post-booster mass	$[kg]$
$m_{support}$	Thrust supporter mass	$[kg]$
m_{wiring}	Wiring mass	$[kg]$
M_f	Final stack mass	$[kg]$
M_F	Fairing mass	$[kg]$
M_o	Initial stack mass	$[kg]$

M_p	Propellant mass	$[kg]$
M_s	Structural mass	$[kg]$
Ma	Mach number	$[-]$
MER	Mass Estimation Relation	$[kg/m^2]$
N	Normal force	$[N]$
q_w	Wind-induced dynamic pressure	$[Pa]$
ρ_0	Air density	$[kg/m^3]$
R_e	Earth radius	$[m]$
S_F	Fairing surface	$[m/s^2]$
t_{coast}	Coasting time	$[s]$
θ	Pitch angle	$[rad]$
$\dot{\theta}$	Angular velocity	$[rad/s]$
$\ddot{\theta}$	Angular acceleration	$[rad/s^2]$
T	Thrust	$[N]$
T_{vacuum}	Thrust in vacuum	$[N]$
V	Volume	$[m^3]$
v_w	Wind-induced velocity	$[m/s]$
$\dot{v}$	Rocket acceleration	$[m/s^2]$

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Chapter 1

Introduction

This study will focus on the preliminary design of an ICBM for civil use, useful for carrying a maximum payload of 100 kg, and subject to a maximum load of 9 g in the re-entry trajectory. A project management chapter will provide a preface to the technical study. In this chapter, the customer's and self-imposed requirements are discussed and the house of quality development. Then the alternatives presented by competitors are considered. The Concept Design chapter will present the main findings and illustrate the process to arrive at them, reached by the iteration of aerodynamics, propulsion, structures, thermal, guidance and trajectory departments. The geometry was derived from the dimensions of the Minutemann III, which was used as a baseline. For the engines, solutions were chosen that were already in the catalogue or in development that have a star grain regression. A 3-D panel method had to be implemented to obtain the pressure distribution, and to verify the stability, and an accurate aerodynamic analysis was developed with Ansys in the case of hypersonic flow, to see the forces the ICBM was subjected to. At the same time, a trajectory team worked out the possible path, which the rocket would have to follow in the three chosen destinations, also associating a guidance and control part to optimise the trajectory and state of the rocket. A part of the team then carried out a detailed analysis of the force distribution and validation of the material to be used as the case. They then analyse the re-entry solutions and thermal loads, choosing a parachute re-entry in which the capsule rotates to ensure less thermal impact; grid fins will also be added for greater control and future development.

1.1 Requirements and Customer's Specification

The requirements established by the customer, which the team endeavored to meet at all costs, are as follows:

- The payload must not endure accelerations exceeding 9 g throughout all mission phases.
- The payload's weight ranges between 10 and 100 kg.
- The time to market is 5 years.
- one-way delivery service from Europe to the USA, which includes establishing three landing sites strategically positioned to cover the East Coast, West Coast, and the Midwest.

- The pricing is not considered a limiting factor in this endeavor.
- The shipment is encased in a capsule designed for secure retrieval, with the possibility of not necessarily reusing it.

As the customer requested a payload mass ranging from 10 to 100 kg and does not impose any restrictions concerning economic factors or the reuse of the capsule, the decision was made to incorporate a dummy mass to stabilize the rocket when the payload weighed less than 100 kg. The initial four requests are mandatory imposed by our client, while the subsequent ones are self-imposed criteria deemed necessary to strike an acceptable balance between performance and programmatic aspects. Requirements:

1. Acceleration: The maximum permissible acceleration experienced by the payload during both ascent and descent phases.
2. Payload Capability: The carrying capacity in terms of mass and volume allocation within the fairing designated for the payload.
3. Time-to-Market: The timeframe required to finalize a detailed design incorporating innovative components, while ensuring compliance with verification and testing regulations across diverse international jurisdictions
4. Safe Disposal: Procedures and protocols implemented to mitigate risks posed to inhabitants in regions traversed by the missile's trajectory. Emphasis is placed on circumventing flight paths over inhabited areas during the initial phase, characterized by higher failure probabilities. Additionally, stages are intentionally released in uninhabited zones with minimal residual hazardous materials.
5. Ground Safety: Safety protocols essential for the loading and positioning of the rocket on the launch pad, prioritizing the safety of ground personnel involved in the operation and ensuring a successful launch.
6. Preparation Time: The duration required to load rocket engines and ready the rocket for departure.
7. Range Flexibility: The missile's capability to modify its trajectory and distance covered, enabling it to target diverse locations.
8. Flight Time: The duration taken for the payload to reach specified destinations, calculated from the moment the payload is integrated into the vehicle positioned on the launch pad.
9. Sub-Orbital flight: It cannot reach orbital velocities.

1.2 House of Quality

The utilization of a House of Quality (HoQ) matrix in the preliminary design process is a helping tool to bridge customer needs with specific engineering parameters. During this phase, the HoQ matrix serves as a foundational tool, enabling the translation of customer requirements, gathered through market research and literature review, into technical specifications crucial for the product design. Additionally, it relates the relationship between each different engineering parameter in the direction of improvement to see what may lower the overall performances. The house of quality file is in the Appendix.

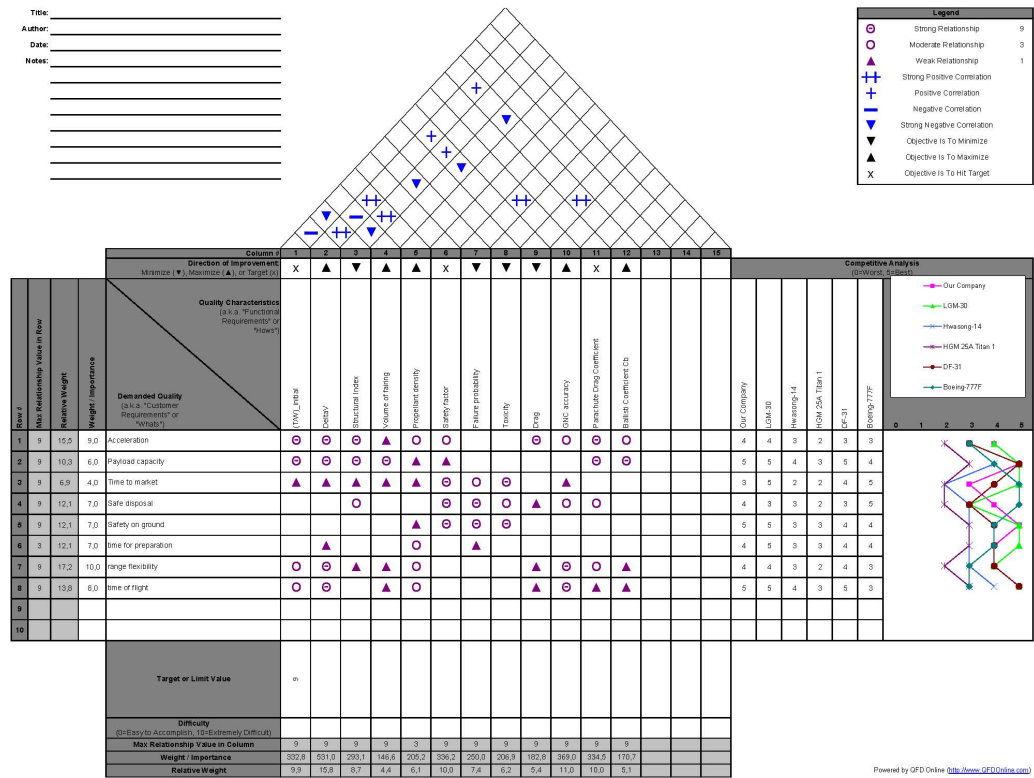


Figure 1.1: House of quality

1.2.1 Competitors

The analysis of competitors presents a comprehensive evaluation of key parameters for several strategic systems, including the Minuteman III serving as the baseline, alongside competitors such as the Boeing 777F, HGM 25A Titan-1, Hwasong-14, and DF-31, in addition to a hypothetical design. Parameters including acceleration, payload capacity, time to market, safe disposal, safety on ground, time of preparation, range flexibility, and time of flight have been meticulously assessed to gauge their comparative performance and operational effectiveness. Each parameter is assessed on a scale ranging from 0 to 5, where higher values signify superior performance. The hypothetical design, alongside competitors, is evaluated across these parameters to provide insights into their relative strengths and weaknesses in strategic ballistic missile capabilities. The table in the Appendix, ?? endeavours to offer a comparative assessment of these systems, emphasizing their distinct performance metrics in crucial domains to facilitate strategic decision-making.

Assessed parameter	Hypothetical design	Minuteman III	Boeing 777F	DF-31	Titan I	Hwasong-14
Acceleration	4	4	3	3	2	3
Payload Capacity	5	5	4	5	3	4
Time to market	3	5	5	4	2	2
Safe disposal	4	3	5	3	2	3
Safety on ground	5	5	4	4	3	3
Time of preparation	4	5	4	4	3	3
Range flexibility	4	4	3	4	2	3
Time of flight	5	5	3	5	3	4
Total	34	36	31	32	20	25

Table 1.1: Possible baseline comparison

1. LGM-30 Minuteman III: Widely acknowledged for its considerable range capacity, the Minuteman III showcases an expansive operational reach, a pivotal attribute that bolsters its strategic significance. Furthermore, stringent safety protocols have been implemented to mitigate potential toxicity risks associated with its propulsion system, thereby ensuring heightened safety measures on the ground.
2. Boeing 777F: While displaying commendable performance across various aspects, its range falls short in comparison to the Minuteman III, constraining its operational reach in extensive long-range strategic deployments. As a civilian cargo aircraft, the Boeing 777F does not inherently pose toxicity risks that could significantly impact ground safety.
3. HGM 25A Titan-1, Hwasong-14, and DF-31: These systems demonstrate a diverse range capabilities, yet none surpasses the extensive range of the Minuteman III. In terms of ground safety, certain older systems like the Titan-1 have historically encountered concerns regarding the toxicity of their propellants, posing potential risks in cases of accidental exposure or mishaps during handling or storage.
4. Hypothetical Design: Building upon the foundational features of the Minuteman III, the hypothetical design aims to exceed or equal its range capabilities. Prioritizing ground safety during the design phase underscores the importance of minimizing potential toxicity risks through advanced engineering and the utilization of safer propulsion systems.

Chapter 2

Project Management

2.1 Functional Tree

The Functional Tree is an essential methodological approach for the design and development of complex systems such as ICBMs. At this early design stage, it is used to examine the development of the vehicle's conceptual mission, identifying its implications in the design. As shown in the appendix, it is seen that the procedure follows a time approach, stating the necessary operations that must be respected to then proceed with the next operations, beginning from the launchpad operations, continuing with the on-flight operations and ending with the payload deployment. This helps to see the criticalities in each phase and what operation may act as a bottleneck.

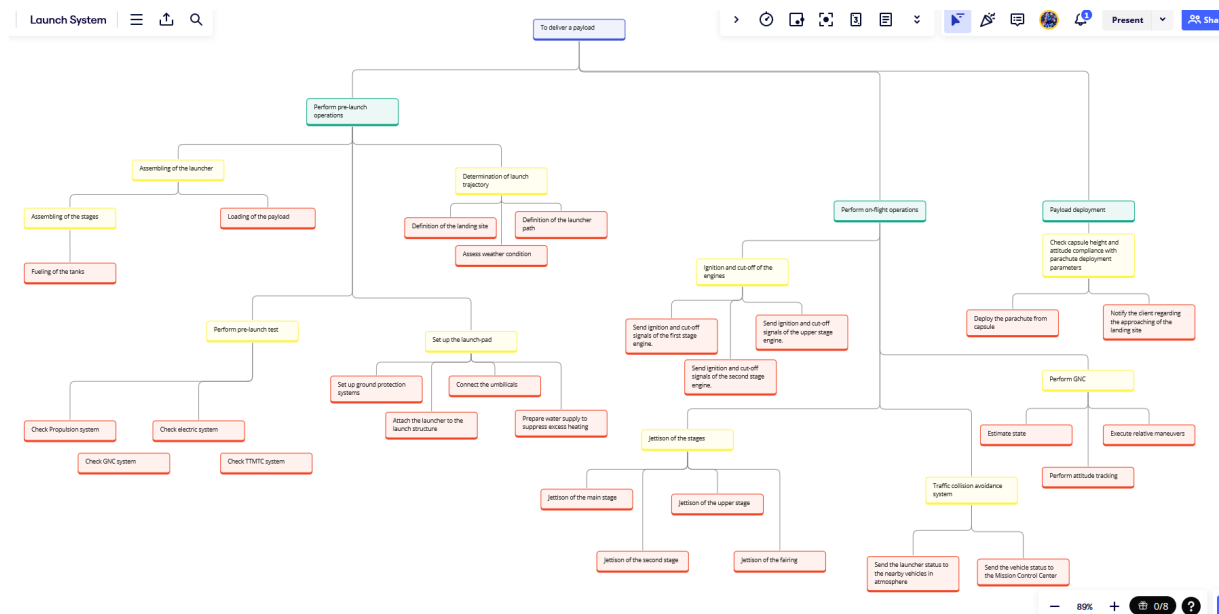


Figure 2.1: In-depth functional tree

2.2 Product Tree

In this application, a product tree is used to represent the various components and subsystems involved in the launch process, breaking down the launcher into its subsystem and

consequently, its constituent parts, to have a better understanding of how each element is made of and interacts. Being this a conceptual design, the focus deepens up to the subsystem level, and a more detailed structural breakdown may be done in later phases of the design. In the appendix, a first draft of the product tree can be consulted to have a better understanding.

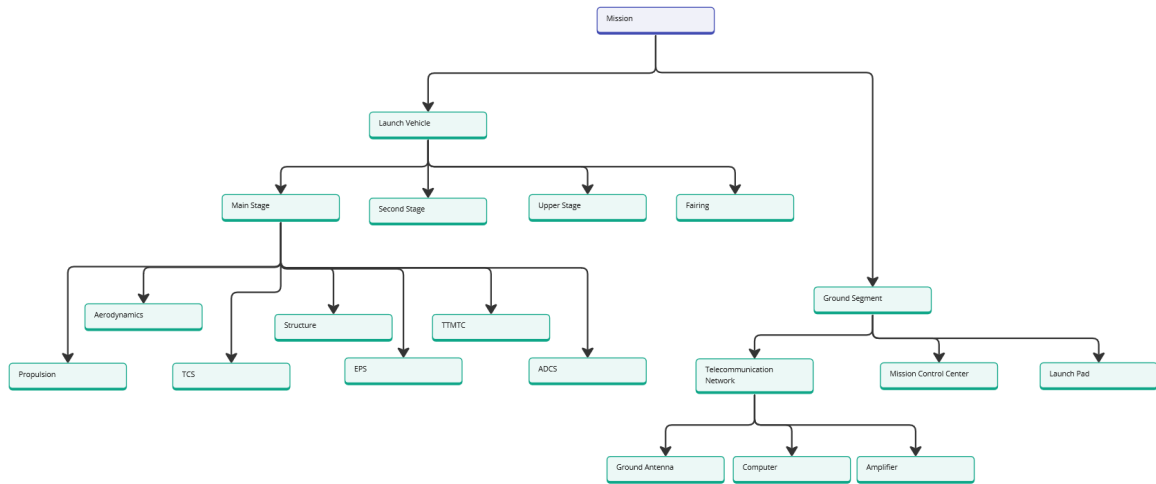


Figure 2.2: Product tree

2.3 Workflow

The figure shows the workflow used to determine this preliminary analysis. It is necessary to start from the analysis of requirements given by the customer, then it is necessary to carry out mainly encyclopedic work, to search for as much information and data as possible in order to study possible competitors and their proposals, and at the same time to be able to find an already developed design, which followed our idea, to use as a baseline. Specifically, a methodical iterative approach is essential, following the precedent set by analogous systems. The work, resulting from the scaling procedure that initiated the concept design phase, is divided according to the skills of the individuals between the Trajectory, Aerodynamics and Propulsion sectors, so that the work could be carried out in parallel as quickly and efficiently as possible, with a continuous exchange of information between the departments. All of the outputs collected by these individual sectors were then fed into the last stage of the iterative process, namely the structural analysis, which allows the study of the distribution of stresses and the consequent choice of material to be used for the case that encompasses the entire length of the ICBM.

The initial iteration distinctly reveals how the output from each stage within the design process seamlessly transitions into the subsequent phase as input. Subsequent to each iteration, an assessment is conducted to ascertain if the specified technical prerequisites have been fulfilled. If the evaluation yields a negative response, it necessitates initiating a fresh iteration to refine the design. Conversely, a positive determination concludes the process. Modifications might impact either the design workflow or the scaling process and consequently the baseline.

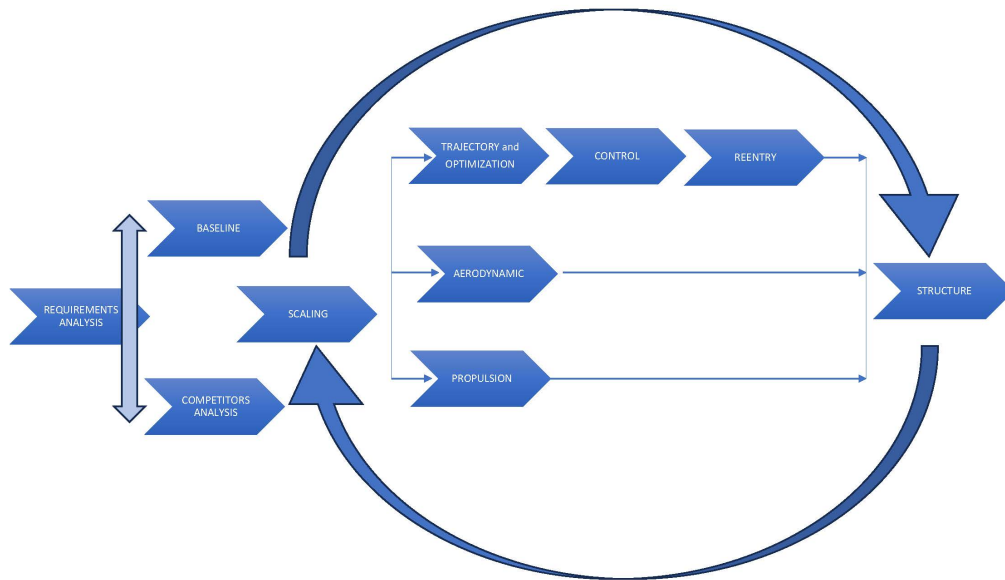


Figure 2.3: Workflow Overview

Chapter 3

Scaling

The scaling, given the new payload mass and the mission profile, aims to design a smaller launcher, obtaining the new masses and dimensions as a result. In order to size a launcher achieving the target mission, the development started from a choice of a launcher with similar performances, the baseline.

3.1 Baseline Selection

The Minuteman III is an intercontinental ballistic missile system developed in 1970 by the United States Air Force (USAF) as part of its strategic nuclear deterrent. It is designed to hold a 1000 kg payload consisting of 3 W62 warheads [14]. The Minuteman III has an operational range of over 10000 km and boasts exceptional accuracy, allowing it to precisely target specific locations.

- Multiple Independently Targetable Re-entry Vehicles (MIRVs): This system is capable of carrying up to three nuclear warheads, each with independent targeting capabilities. MIRVs enhance the missile's effectiveness by allowing it to strike multiple targets with a single launch.
- Quick Response Time: The Minuteman III is designed for rapid launch, allowing the USAF to respond swiftly to potential threats, contributing to deterrence strategies.
- Reliability and Readiness: The system maintains high levels of readiness, with regular maintenance and testing ensuring operational reliability.
- Solid-Fuel Propulsion: the Minuteman III employs solid-fuel propulsion, enabling quicker launch times, increased reliability, and reduced maintenance requirements. It is a four-stage solid-propellant intercontinental ballistic missile:

Some features of the Minuteman III did not meet the project requirements, namely:

1. The payload is 10 times larger
2. A post-boost is not foreseen in the final configuration, as in our case the re-entry will be performed with an injection parachute and grid fin
3. Having only one deliverable, to be released in one area, it was not necessary to use re-entry vehicles

Stage [–]	Name of motor [–]	T [kN]	I_s [s]
Stage 1	Thiokol M55 solid-fuel rocket	792	262
Stage 2	Aerojet General SR19-AJ-1 solid-fuel rocket	268	288
Stage 3	Aerojet-Thiokol SR73-AJ-TC-1 solid-fuel rocket	152	285
Post-booster	Rocketdyne RS-14 liquid-fuel rocket	1.39	315

Table 3.1: Baseline data

	Length [m]	Diameter [m]	Mass [kg]
Stage 1	7.49	1.67	23077
Stage 2	4.10	1.31	7032
Stage 3	2.30	1.30	3600
PB	0.76	1.30	471.07
Fairing	3.62	1.30	1020
Total	18.3	1.67	36030

Table 3.2: Baseline data [10]

4. The engines selected were not designed to be placed in succession, but were chosen from a catalogue[3] differently to the Minuteman ones, which were designed to be placed in sequence and used together.

3.2 Scaling

Since the new rocket’s application is similar to the baseline, the Δv profile is supposed to be the same for both units and it is computed with Tsiolkovsky’s formula:

$$\Delta v = \sum_{i=1}^3 I_{s_i} \cdot g_0 \cdot \ln\left(\frac{M_{oi}}{M_{fi}}\right) \quad (3.1)$$

By using this formula, the drag and gravity losses are already taken into account because the minuteman is designed with a Δv ensuring that it reaches its target despite the losses. From this point, the scaling procedure begins from the fairing and progresses up to the first stage, since the sizing variable is the new payload, the modifications start from the top. The payload mass is considered as the upper value of the mass range, to guarantee the capability of delivery in the worst loading condition. In the first iterations, the fairing shape is initially considered as a cone for simpler calculations, then it is modeled as an ogive shape coming from the aerodynamics computations. The fairing volume is computed by shrinking the baseline by the payload ratio. The diameter of the fairing is constrained to $d_f = 0.7\text{m}$ due to the smaller payload volume, which allows the diameter to be reduced and makes the launch vehicle more slender. The mass is defined by using the mass estimation relation (MER), a ratio of the mass and the surface necessary to ensure good performance. The same MER is applied to the post-booster mass. This item is used for the control of the reentry vehicle, which at first was modeled as an added minor stage, but due to the different functionality compared to a stage, it was considered as an inert

mass. This contains all the control mass, such as avionics and wiring.

After that, the scaling for the three stages is initialized, where each shape is modeled as cylindrical for simplicity. In this analysis, ϵ_s and I_s are kept equal to the baseline data, considering keeping the same I_s performances even for a smaller motor by having the same propellant grain material and a linear mass reduction as an approximation. The scaling proceeds by applying the Lagrangian method for the optimal staging [9]: this method allows to obtain a Lagrangian coefficient able to determine the stage masses repartitioning to minimize the liftoff mass, not accounting for losses, but as stated previously, it is taken into account in the final design of the Minuteman.

$$\Delta v = \sum_{i=1}^3 I_{s_i} \cdot g_0 \cdot \ln\left(\frac{I_{s_i} \cdot g_0 \cdot \lambda - 1}{\epsilon_{s_i} \cdot I_{s_i} \cdot g_0 \cdot \lambda}\right) \quad (3.2)$$

A two stages optimal staging approach was considered but was discarded since the optimization showed a strong reduction in performance: fineness ratio was greatly reduced, obtaining a more blunt body causing a higher drag, and the accelerations exceeded the limit value of 9g. Due to these considerations, a three stages optimal staging approach was chosen.

Once the mass have been sized, the dimensions (l,d,V) are obtained. The diameters of the fairing, and the third and second stage are maintained equal to the assumed d_F but the first stage diameter is increased, to be consistent with the baseline shape. Subsequently, with this assumptions, the volumes are derived by the baseline missile density: the ratio between stack mass and volume of the baseline is obtained and is assumed constant to define the new volume of the scaled launch vehicle. Finally, l/d is computed by considering the full length and a weighted average diameter with the lengths.

As the theoretical staging got to an end, to validate the model, existing stages are chosen from the Northrop-Grumman catalogue, to provide a propulsive system able to satisfy the time-to-market requirement, obtaining the final rocket data. The focus is made on the diameter compliance, since each upper stage must have a smaller diameter than the lower one, on the I_{tot} to verify that the propulsion unit can obtain similar performances with the estimated calculations, and on the production status where the units chosen are all flight-proven and can be utilized in five years.

Finally, from this point, more detailed structural mass such as wiring and thrust supporter masses are computed following similar estimation procedures as the previous process.

3.3 Scaling Process

First of all the fairing and post-booster parameters are designed starting from the volume obtained with the payload ratio 3.3:

$$V_{F,pb} = V_{F,pb,base} \cdot \frac{M_{pay}}{M_{pay,base}} \quad (3.3)$$

After that, the masses consider a MER imposed at 13.3 kg/m^2 to get the masses:

$$M_{F,pb} = S_{F,pb} \cdot MER \quad (3.4)$$

The avionics mass, part of the post-booster mass, is considered to be 80 percent allocated in it and 20 percent in the lower stages because it is the main core of the control system,

additionally, the value of 75kg is taken considering the threshold of 1000kg [7] of payload mass:

$$M_{avio,pb} = 75 \cdot 0.8 \quad (3.5)$$

The inert mass fractions are derived from the baseline masses and are used to compute scaled motor masses for the optimal staging. It is seen that the second stage has a bigger value, following the trend of propulsion units of smaller sizes, while the first stage has a structural index exceeding the recommended range of 0.05-0.08 but it is expected due to the high thrust-over-weight ratio at launch causing higher stresses. The following equations are used to perform the optimal staging:

$$\epsilon_{si} = \frac{M_{si}}{M_{si} + M_{pi}} \quad (3.6)$$

$$M_i = \frac{n_i - 1}{1 - \epsilon_{si} \cdot n_i} \cdot (M_F + M_{pay} + M_{pb} + M_{i-1}) \quad (3.7)$$

$$n_i = \frac{I_{si} \cdot g_0 \cdot \lambda - 1}{\epsilon_{si} \cdot I_{si} \cdot g_0 \cdot \lambda} \quad (3.8)$$

The volume of each stage is obtained by the missile density computed for every stage by using a fitting law coming from historical data, as shown in Appendix:

$$\rho_{base} = \frac{M_{0,base}}{V_{0,base}} \quad (3.9)$$

Then, each respective length and diameter is retrieved and fineness ratio is derived, computing the average diameter considering each stage, fairing and the other inert masses as follows:

$$d_{average} = \frac{\sum_{i=1}^5 l_i \cdot d_i}{l} \quad (3.10)$$

$$l/d = \frac{l}{d_{average}} \quad (3.11)$$

It shall be noticed that all the dimensions of the designed system are smaller than the Minuteman III's ones. This fulfill our expectations since the concept of the analysis is to reduce the baseline size to achieve the target mission. The choice of compatible motors from the manufacturer catalogue lead to slightly bigger radius to satisfy the performance requirements. In order to maintain a finesse ratio L/D sufficiently close to the Minuteman's one, a longer fairing is designed. Finally:

Minuteman III	TBD Theoretical	TBD Catalogue Motor
13.354	9.254	12.656

Thus, increasing the fairing length we achieved the desider L/D .

3.4 Choice of motors

Since most launch failures are due to a failure[9] in the propulsion system, it was decided to use engines with an advanced TRL. It was also chosen to use solid propellants only, in order to significantly reduce the risk of release of dangerous substances on the launchpad and to speed up the loading process, as the propellant is already integrated. The system

can be then stored as a *ready to launch* system.

As the existing motors were evaluated from a catalog [3], the final properties differ because it assembles three pre-existing units and some shifting from the optimum had to be expected. The first motor chosen is the *Orion 50XL*: although the I_{tot} is quite higher than the desired one, it is the option that gives the most similar compliance with the dimensions, the thrust profile, I_s and M_0 concerning the scaling outputs. This motor has a QDL-1, HTPB POLYMER, 19% aluminum propellant grain initially used for the Pegasus XL mission, including a movable nozzle for thrust vector control of $\pm 5^\circ$ vector capability. Other options were initially considered, such as the *ASAS 28-185/185V TE-T-1032* and *ASAS 21-120 TE-M-1059-1* [3]. These alternatives got discarded due to the lower values of I_{tot} compared to the scaling outputs and due to smaller diameters than the existing motors chosen for the other stages to obtain good performances. The second and third motors are part of the STAR motor series, mainly used for the apogee kick maneuvers, have an elliptical tank shape and use a titanium case. They are the *STAR 37FM TE-M-783* for the second stage and the *STAR 30BP TE-M-700-20* for the third stage; both flight proven. Other possible options such as *STAR 37FMV TE-M-1139* were discarded because they are currently under development, so they are not compliant with the TRL requirement. To account for the three different diameters, the interstages were added as truncated cones and not as cylinders to make the connections between the stages and have a linear variation in the diameter. Once the motors were obtained, the nose was remodeled as an ogive, which was the final designed shape, that guarantees a higher volume for the payload storage. Additionally, length and diameter were raised and decreased respectively to account for the doubled volume compared to the nose cone to favor the fineness ratio increase. Furthermore, the precise data for the structural masses were derived: avionics mass stays the same, since payload does not change, wiring mass changes since the length varies and similarly, the thrust supporters mass varies with T[11]:

$$m_{wiring} = 1.43 \cdot l_i \quad (3.12)$$

$$m_{support} = 2.55T \cdot 10^{-4} \quad (3.13)$$

The output performances change, the fineness ratio gets reduced to 9.0268 because of the non-compliance with the diameters and the final I_{tot} of each stage was not the optimal value. The final results can be consulted below:

	Motor	Thrust [N]	I_s [s]	Δv [m/s]	ϵ_s [—]
Stage 1	Orion 50XL	157960.79	265.48	2651.84	0.0909
Stage 2	STAR 37FM TE-M-783	48160.90	268.14	2313.39	0.0712
Stage 3	STAR 30BP TE-M-700-20	26622.61	269.98	3659.73	0.0694

Table 3.3: Performance results of new launch vehicle

	Length [m]	Diameter [m]	Mass [kg]
Stage 1	3.0709	1.2751	4306.4
Stage 2	1.6891	0.9347	1148.0
Stage 3	1.5062	0.7620	542.8
PB (average)	0.2598	0.7620	115.19
Fairing+payload	1.7400	0.7020	131.69
Total	8.8104	0.9157	6128.9

Table 3.4: Mass and dimensions results of new launch vehicle

	M_s [kg]	M_p [kg]	m_{wiring} [kg]	$m_{support}$ [kg]
Stage 1	391.45	3914.96	5.345	40.28
Stage 2	81.69	1065.99	3.540	12.28
Stage 3	37.69	505.12	1.557	6.79
PB (average)	115.19	-	0.375	-

Table 3.5: Secondary masses of new launch vehicle

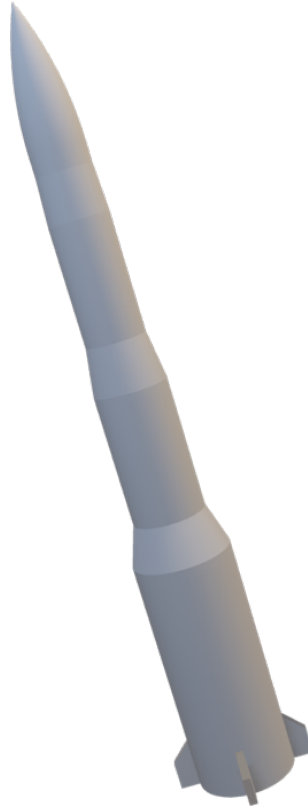


Figure 3.1: 3D render of the developed system

Further Tables and Diagrams

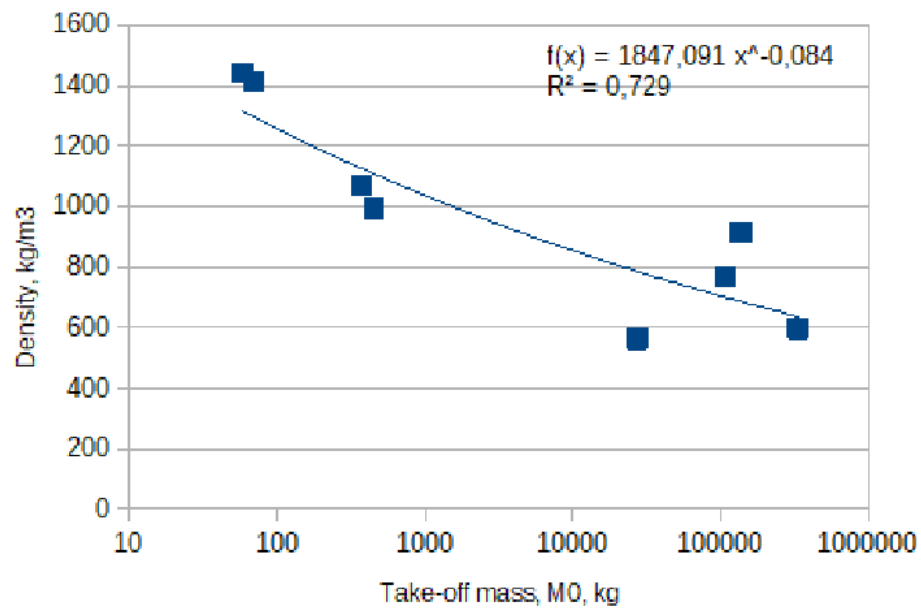


Figure 3.2: Missile density fitting law

Material	σ_{yeld}	E
[—]	[GPa]	[MPa]
Titanium	1030	113

Table 3.6: Titanium alloy properties [6]



Figure 3.3: 3D render of the developed system

Chapter 4

Aerodynamics

4.1 Hypersonic regime

4.1.1 Newtonian theory

The Newtonian theory [12] in aerodynamics is a fundamental approach used in the study of hypersonic flight mechanics, particularly for spacecraft and re-entry vehicles. This theory is based on the concept of the shock layer being extremely thin, making the shock practically coincide with the body surface. It allows for the subdivision of a body's surface into panels, each treated separately to determine the pressure force acting on it. The total force and moment structure of a vehicle can be computed by summing the contributions from each surface element.

The distribution of forces along a rocket's body enables the calculation of key aerodynamic parameters such as normal forces, axial forces, lift, and drag. Crucially, it allows for the determination of the center of pressure—a vital aspect in rocket aerodynamics. The center of pressure's position is crucial for evaluating the rocket's stability, particularly when juxtaposed with its center of gravity. Typically, computing the center of pressure is a complex task. However, leveraging the force distribution insights provided by Newton's theory simplifies this process. We can employ the following equation [2]:

$$cp = \frac{\int xp(x) dx}{\int p(x) dx} \quad (4.1)$$

Our geometry being divided into panels, this process is done numerically by taking the sum of the value of each panel times the distance of the centroid with respect to a reference point, in our case the nose of the rocket.

A possible improvement to the Newton method can be made using Prandtl-Meyer expansion, which is a phenomenon observed in supersonic flows when they turn around a convex corner. This process forms what is known as a Prandtl-Meyer expansion fan, which consists of an infinite number of Mach waves that diverge from the corner point.

4.1.2 Limitations

The Newtonian theory is mostly suited for very high Mach numbers, in fact, it reduces in accuracy in the supersonic regime ($Ma < 4$). We will nonetheless assume that it is

accurate enough for our conceptual design.

4.1.3 Results

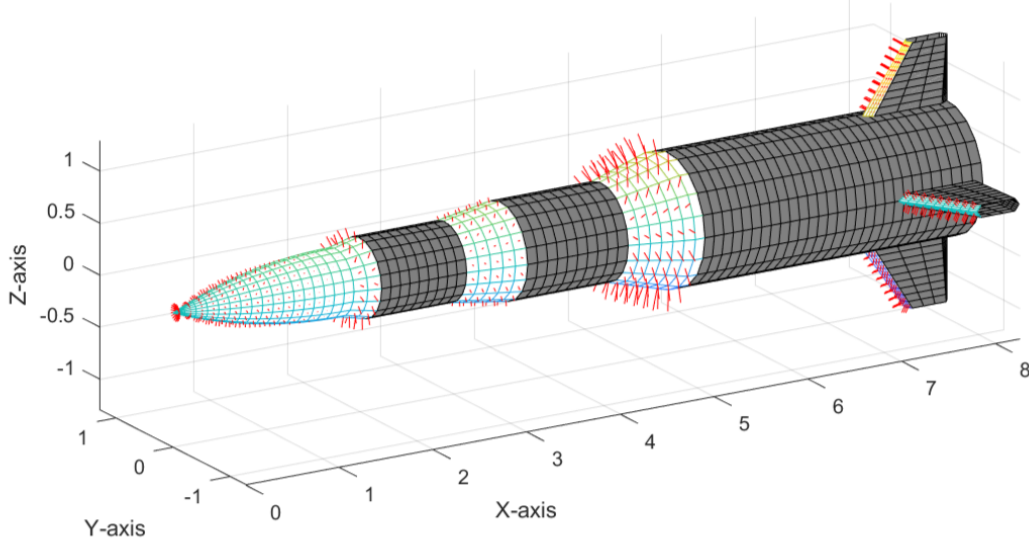


Figure 4.1: 3D view of the discretized rocket

After a successful implementation of the modified Newtonian method, we obtain the following forces distribution. The conditions seen here correspond to the Max-q conditions that were calculated. We can see the manifestation of the theory principles such as null values of C_p for panels that are parallel to the velocity (grey mesh) as well as the ones in the shadow region.

The tip of the rocket's nose was modeled as a sharp cone for the first set of panels, applying the following equation from Newton's theory:

$$C_{p,N,c} = \frac{2 \sin^2 \delta_c}{\left(1 - \frac{\varepsilon}{4}\right) \cos^2(\theta - \delta_c)} \quad (4.2)$$

Where δ_c is the half-angle of the cone, θ the angle between the local normal unit vector and the velocity, and ε the ratio of density upstream of a shock to that downstream of a shock.

As for the fins, if we were mainly in a subsonic regime, a rounded leading edge and a wedge shaped trailing edge would be suitable, but since we spend most our time inside a supersonic and hypersonic flow, the leading edge and trailing edge should both be wedge shaped. Another interesting possibility that could be further studied is the unsymmetrical airfoil where the leading edge and trailing edge are chisel-shaped, such that greater lift is developed on one surface of the fin. This would induce a slight rotation along the longitudinal axis that would be beneficial for the stability, this particular profile was used on the Cirrus One rocket with great results [4].

Since our leading edge is modeled as a wedge, the Newtonian theory requires us to apply this slightly different equation:

$$C_{p,N,w} = \frac{2 \sin^2 \delta}{(1 - \varepsilon) \cos^2(\theta - \delta)} \quad (4.3)$$

Using equation (4.3), we can compute the center of pressure and make sure it is aft to the center of gravity. The green star represents the position of the center of pressure while considering a body with 4 fins. The thickness of the fins as well as the half-angle of the wedge have a very considerable effect on the position of the center of pressure, but it is important to find a trade-off between the drag and the stability generated.

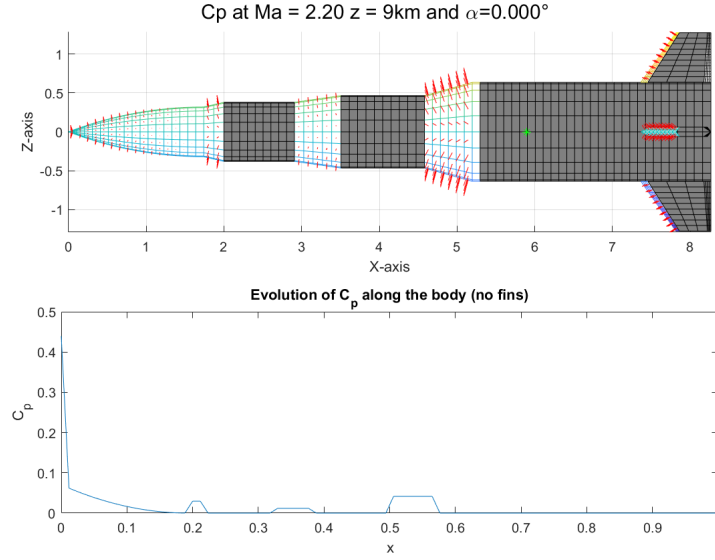


Figure 4.2: Pressure distribution

4.1.4 Evolution of the center of pressure

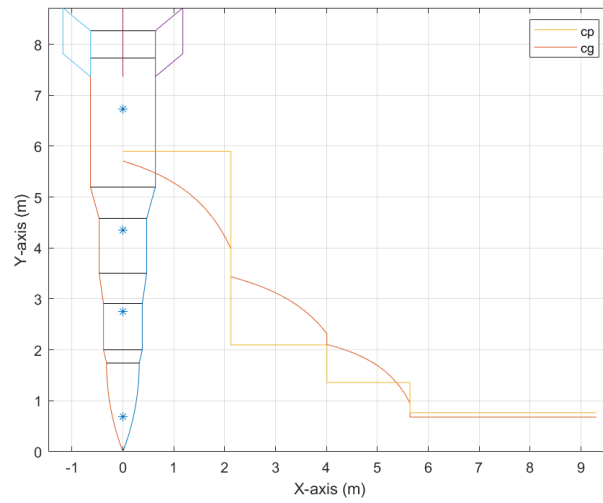


Figure 4.3: Evolution of the position of cg and theoretical cp throughout the path

If we superpose the evolution of the positions of the center of pressure and center of gravity we can obtain a clear representation of when the center of pressure is not aft to the center of gravity. The first stack and the payload stage stay stable along their entire respective phases. The third stack (including the third stage and the payload) may seem unstable but it will go beyond the Karman line so we can neglect all aerodynamic effects, which leaves us to the second stack which is unstable throughout the whole phase. To remedy this problem, we can consider stabilization through thrust vectoring and if required, the implementation of grid fins in a X-wing configuration.

4.2 Subsonic regime

Although most of the flight path happens at supersonic and hypersonic regimes, it can still be useful to compute the center of pressure of our rocket in subsonic flow.

To achieve this, instead of using a paneling method like Newton's theory, we will use another popular methodology known as component buildup. It consists of dividing the rocket into multiple components in order to compute the properties of each separately and then sum up their contributions.

The Barrowman equations [1] are a powerful tool used for preliminary designs as they are extremely simple to implement and test for subsonic regimes (the accuracy is heavily affected in supersonic flows). They allow us to calculate the position of the center of pressure following a component build-up method.

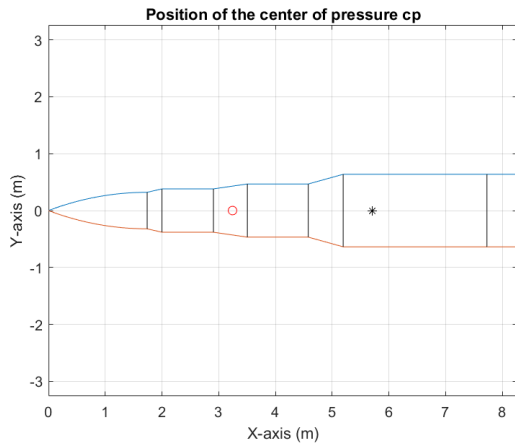


Figure 4.4: Position of cp with no fins

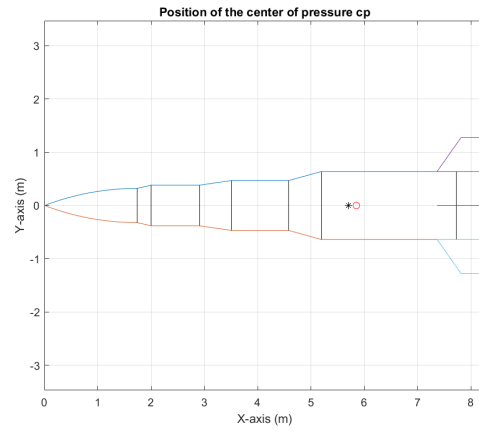


Figure 4.5: Position of cp with 4 fins

We can notice on the figures given above, the position of the center of pressure (red circle) with respect to the center of gravity (black star) and appreciate the importance and effect of a set of fins on the stability of the rocket as they pull back the center of pressure to be aft of the center of gravity, thus achieving stability for the full stack. These values were also verified using the software OpenRocket with very close results.

4.3 Re-entry

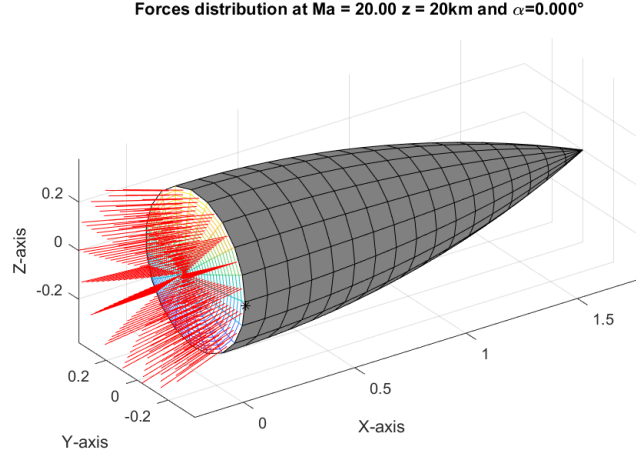


Figure 4.6: Pressure distribution during re-entry

The re-entry phase starts at $Ma \approx 25$ and steadily decreases as we lose altitude but most of it stays in the hypersonic regime. Thus, we can use the Newtonian theory to visualize the forces distribution, calculate the drag force being applied at a certain point, and use it to find C_d . If we apply the method at 20km which coincidentally corresponds to a Mach number of 20 according to our trajectory calculation, we obtain a drag force of $D = 1015.8\text{kN}$. Using equation 4.4, we can find a drag coefficient of $C_d = 1.7$. The values of density were obtained using the 1976 Standard Atmosphere Calculator developed by Brent Lewis.

$$F_d = \frac{1}{2} \rho_0 v^2 C_d A \quad (4.4)$$

4.4 CFD

With the defined geometry from the scaling, and as both a complement and a check of the results obtained previously with the empirical, Barrowman and Newtonian methods, CFD simulations were carried out with Ansys Fluent. These simulations were done for the first stack geometry, as the aerodynamic forces after the first stage jettison are almost negligible.

Regarding the nozzle exhaust pressure, some simulations were carried out considering a pressure inlet with an area smaller than the one of the stage 1 and interstage 1, as it would be the case with the Orion 50 XL. But as the entire interstage area was considered as the exhaust area, simulations without the nozzle effect, as in unpowered flight, seemed more suitable, as the error computing the base as a pressure inlet would be higher than neglecting the base drag after simulating the missile as in unpowered regime. Therefore, the base drag was computed and subtracted to the stage 1 drag, as it can be considered negligible based on the following equations [8].

$$\begin{cases} C_{D_{base}} = (1 - \frac{A_e}{A_{ref}})(0.12 + 0.13Ma^2) & \text{for } Ma < 1 \\ C_{D_{base}} = (1 - \frac{A_e}{A_{ref}})\frac{0.25}{Ma} & \text{for } Ma > 1 \end{cases} \quad (4.5)$$

4.4.1 Process

For the simulations, the CAD model was created using CATIA V5 and exported as an .stp file. In the Ansys Workbench, the different toolboxes to use were integrated.

Firstly, the CAD model was imported in the Design Modeler where a box was designed around the missile as the fluid control volume. The missile body was subtracted from this control volume with a Boolean operation. Then, the inlet, outlet, volume walls and the different stages surfaces were defined as named surfaces.

In the Mesh toolbox, a tetrahedral cells mesh was generated, with an 8-layer inflation refinement on the walls of the missile, trying to reduce the Y^+ value as much as possible. A Face Sizing was also applied on the walls of the missile, with a slow size transition from them to the control volume walls. Lastly, the mesh was imported into Fluent, where all the Setup and boundary conditions were defined.

For the solver, the energy equation was activated, to work with the Navier-Stokes equations, as the compressibility effects are relevant; the SST k- ω was selected as the viscous model selected for turbulence, with Viscous Heating, Production Limiter and Curvature Correction.

For the boundary conditions, the control domain walls were identified as pressure far fields, introducing for each of the Mach number, the Gauge pressure (computed with the isentropic flow equations from the static pressure) and the temperature were defined.

Finally, the reference area for the drag or lift coefficient was selected and all the reports and plots desired, regarding forces, coefficients or parameters, were specified.

4.4.2 Drag and Lift Coefficients

In the first place, for many Mach number values, the drag and lift coefficients were calculated.

In 4.7, we can observe the total c_{D_0} , or drag coefficient at zero angle of attack, and the c_{D_0} of all three stages, post-burner and nose. The values for the initial calculations considering the base drag are also presented in 4.8.

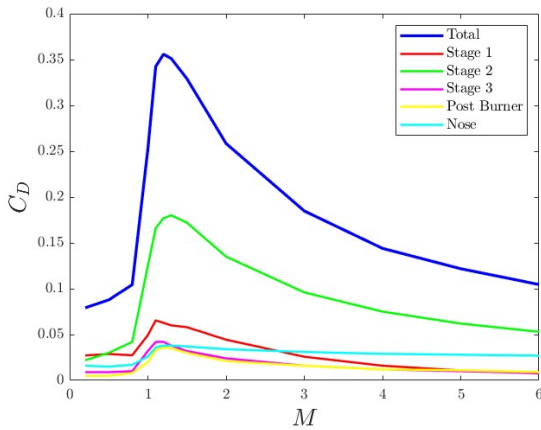


Figure 4.7: C_D contributions (Powered)

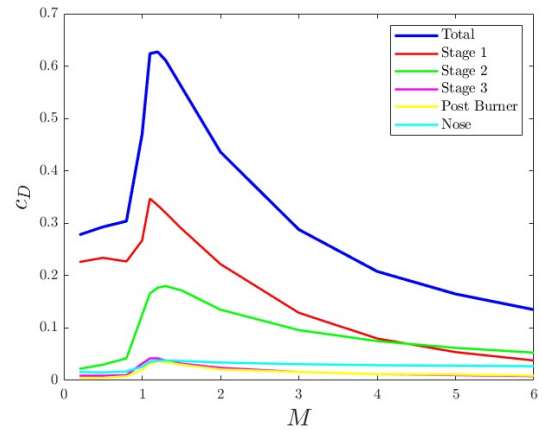


Figure 4.8: C_D contributions (Unpowered)

It can be observed that the values are close to the ones used for the scaling when the nozzle exhaust pressure is considered, as happens in powered flight, and the base coefficient discarded. The values are not high for a first geometry coming directly from the scaling and without any aerodynamic optimization considerations and on par with many other missiles with similar characteristics.

For the drag in the first stage is the most important one in the subsonic region due to the higher importance in this regime of the skin drag. In the case of the second stage, specifically the second inter-stage conical section, the drag is high due to the notable diameter increase from the second stage to the first stage. This implies the appearance of shock-waves in supersonic regime. The diameter increase is much lower in the third stage and post-burner, where the transition is smooth and therefore the drag contribution small.

Lastly, the nose drag is small due to a very aerodynamic ogival shape, but as with every nose its drag contribution increases significantly when the missile gets closer to hypersonic Mach numbers, becoming the second major drag contribution.

Lastly, in 4.9 and 4.10, the drag and lift coefficients with 0, 5 and 10° of angle of attack are presented.

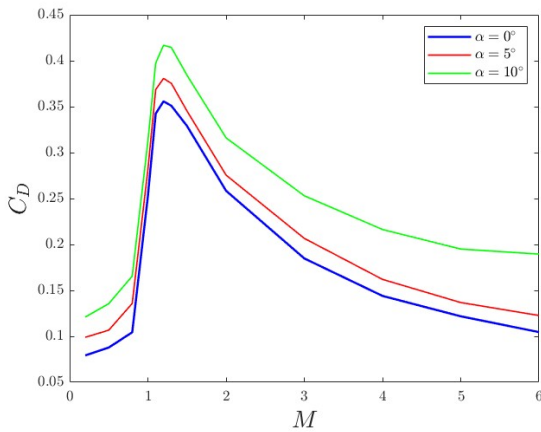


Figure 4.9: C_D evolution for three angles

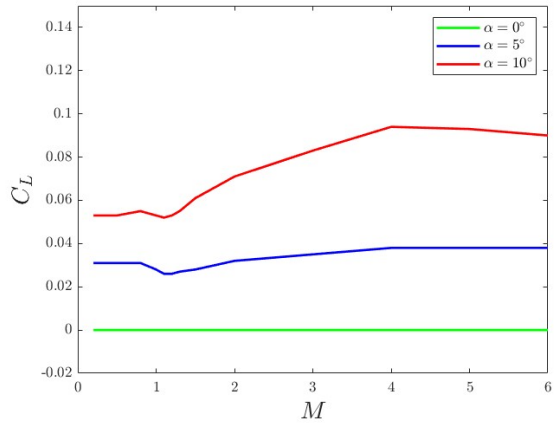


Figure 4.10: C_L evolution for three angles

4.4.3 Center of Pressure position

From the computation of the aerodynamic forces on every section of the missile, the coefficient of pressure position was studied for supersonic and hypersonic regimes, in an attempt of describing the already computed, considered fixed, position more precisely. Many simulations with Mach-pressure conditions from many flight path points were carried out. The temperature variations were not considered, as the effect was negligible. For the rest of the conditions and set-up, the same ones as for the drag and lift coefficient computations were considered. It is presented here the comparison between the fixed value computed previously, before the addition of fins due to stability problems, and the one computed in Fluent.

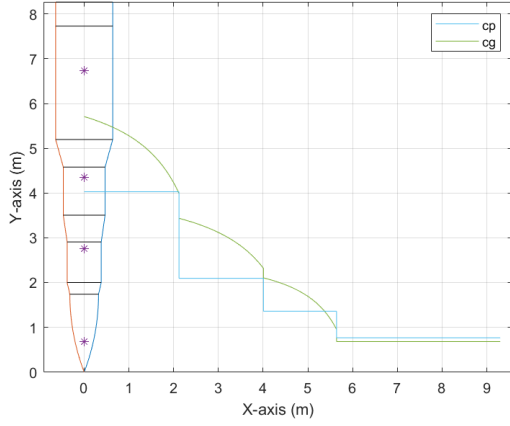


Figure 4.11: Position of C_p for every stack

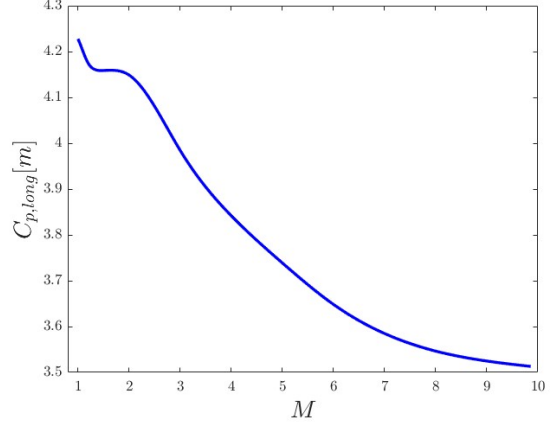


Figure 4.12: C_p position for Stack 1

In 4.12, it can be observed that it exists a tendency of the center of pressure moving afterwards, closer to the base of the missile, as the Mach number increases. These results are congruent with the bibliography [5]. The previously computed in 4.1 is very close to the values in here, which range from around 4.2 to 3.5 meters. This validates the code computing the center of pressure position.

4.4.4 Max-q point forces

Lastly, the aerodynamic forces at the max-q point were computed for its usage in both the structural and stability sections. In the following image, the velocity contour at the maximum dynamic pressure point is presented. The shock waves are visible, as well the high re-circulation due to the non existence of a pressure inlet simulating the exhaust at the missile base. The inclination of the missile with respect to the horizontal plane is the same as in this flight point (max-q).

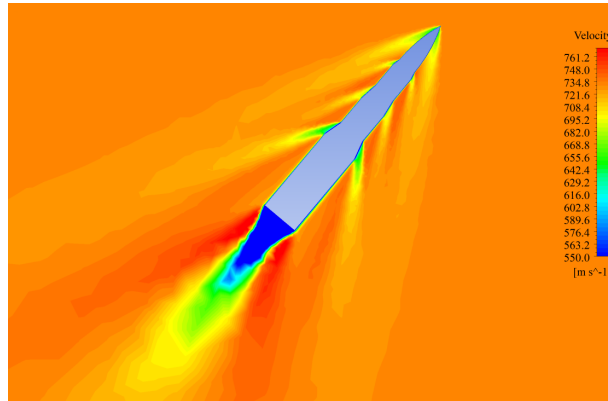


Figure 4.13: Velocity Contour of the missile at the max-q point

Chapter 5

Trajectory

5.1 Trajectory Overview

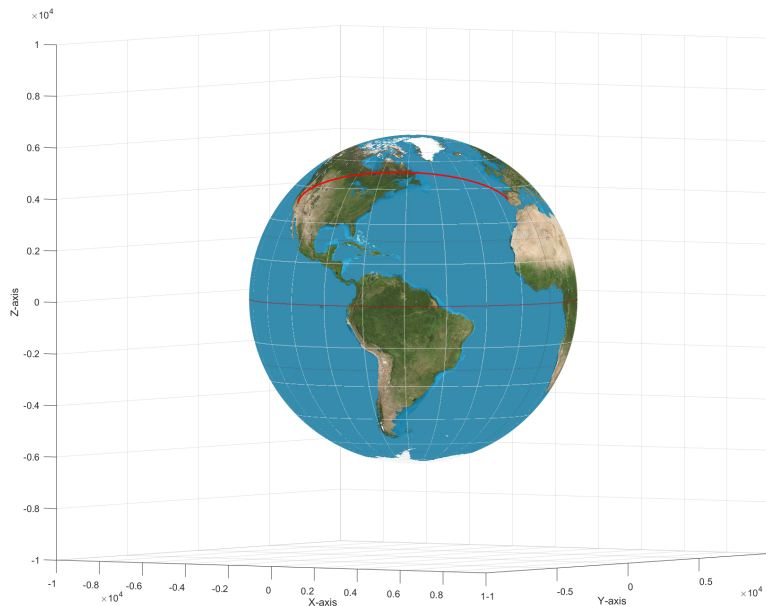


Figure 5.1: 3D Overview of the Trajectory

It is decided to locate the launch site at the Santarem general aviation airport, approximately 70 km from the city of Lisbon. This is a hilly area with little wind, far from the population centres, the nearest one (70000 inhabitants) is to the north and consequently will not be flown over. There are no military bases in the vicinity that can not be flown over. Moreover it is possible to use the facilities and infrastructures already present at this small airport, without requiring permits from the Portuguese state to build new infrastructure. The three planned arrival sites will be in the vicinity of New York, Denver and Sacramento. The most complex pick-up site to choose was the one in New York State because unlike the other two, which have large uninhabited areas outside the city, this one has a high density of population. It was therefore decided to move the site 100 km north of the city of New York, to Montebello (7000 inhabitants), in Sullivan County, where there are unpopulated areas and the terrain is predominantly hilly.

5.2 Trajectory Analysis

5.2.1 Trajectory Model

The trajectory analysis aimed to determine all the physical parameters describing, in a first approximation, the motion of the missile modeled as a point mass. Both the powered and unpowered phases are considered, with the possibility of tuning the coasting time before the jettisoning of the stage. After the optimal analysis, the latter is left apart since optimal coasting time is found to be $t_{\text{coast}} = 0$ s. The simulation is carried according to the following assumption:

- A bi-dimensional reference system passing through the center of mass of the planet is considered;
- Curved and fixed Earth's surface;
- No disturbance along the flight path;
- Thrust aligned with the trajectory.

The last two strong assumptions are *a posteriori* confirmed thanks to a proper control system which permits the missile to follow the trajectory almost always tangentially, with only small angles of thrust vectoring leading to no significant change in the thrust, apart from the ones coming from the thrust profile.

Moreover, three more assumptions are done regarding the mission profile:

- $T \approx T_{\text{vacuum}} \quad \forall t$
- Flight starts vertically and then tilts to the optimal trajectory angle once reached a given optimal altitude;
- After the first steering only a gravity turn is performed;
- Any other steering is only used to achieve the tangent condition between thrust and trajectory. Further study shows that since the steering angles are small, neglecting their effect on the parallel thrust magnitude is justified.

The trajectory code is implemented according to the equation of motion of the rocket given as:

$$\dot{\gamma} = -\frac{1}{v} \left[g - \frac{v^2}{R_e + h} \right] \cos \gamma \quad (5.1)$$

$$\dot{x} = \frac{R_e}{R_e + h} v \cos \gamma \quad (5.2)$$

$$\dot{h} = v \sin \gamma \quad (5.3)$$

$$\dot{v} = \frac{T}{m} - \frac{D}{m} - g \sin \gamma \quad (5.4)$$

Where the curvature of the Earth is considered through the term $\frac{v}{R_e+h} \cos \gamma$.

The system of ordinary differential equation above is then solved using the solver ODE45 built-in in MATLAB and setting as stopping condition the reaching of the ground. Parameters such as height, range, velocity and longitudinal acceleration are extracted and from the latter it is observed a considerable instability of the solution in the first and second stages. The acceleration plots express a significant oscillation at a high frequency that could be explained by the sensitivity of gravity turn to the instantaneous conditions.

5.2.2 Drag Model

During the first iterations, in the starting phase of the project a drag estimation was required. Being the missile object of this study based on an ICBM, the *LGM-30 Minuteman III*, our expertise suggested exploiting another well-known ballistic missile to produce a preliminary model of the drag forces. Given the large amount of information available and the geometrical similarities between the two, the V-2 missile is chosen as baseline for the drag.

Cd	0.25	0.18	0.28	0.36	0.26	0.17	0.15	0.14	0.12	0.11	0.1
M	0	0.5	1.0	1.2	1.5	2.0	2.5	3.0	3.5	4.0	5.0

Table 5.1: Drag & Mach values for the V-2 Missile [10]

In order to compute the Mach number along the trajectory was used the built-in function *atmoscoesa*, exploiting the 1976 COESA model. This model is used up to 84852 m to produce the values for the *speed of sound* a ; after this threshold, the model is no more reliable but, considering the very small density at that height the drag can be neglected after such value.

5.2.3 Thermal Load Analysis in the Trajectory

The trajectory simulation also computes an initial estimation of the thermal load. After the first iteration is highlighted that the major thermal stress occurs in the reentry phase as expected, thus a bottom-down reentry configuration is selected to carry out further simulations. The latter is such to exploit a higher curvature radius of the thermal shielding and thus reducing the thermal load according to the equation 5.5

$$q_{t,c} = C \sqrt{\rho} v^3, \quad C = \left(1.7415 \cdot 10^{-4} \sqrt{\frac{kg}{m}} \right) \frac{1}{\sqrt{R_N}} \quad (5.5)$$

The equation above is meant for the rough estimation of the thermal loads onto a capsule during its reentry, thus applied in the first part of the trajectory, with the nose pointing up, it may lead to some inaccuracy, since the geometry is significantly different from the one of a capsule. In the reentry phase, instead, assuming the bottom down configuration and exposing the curved bottom of the capsule to the airflow, the geometry resembles a capsule-like configuration leading very likely to a better accuracy.

5.3 Trajectory Optimization

The trajectory optimization was carried in parallel with two different approaches: a *brute force approach* and a *Lagrangian multiplier λ approach*. The MATLAB implementation of the brute force approach was carried through nested *while* cycle and *if* statements to account for the constraint on the dynamic pressure and the thermal load, while the two free parameter were γ_0 and h_{turn} .

The *Lagrangian multiplier problem* was solved using the **Matlab** solver **fmincon** with the direct simple shooting method for the three ranges, by giving as input the objective function to minimize and the constraint function. The objective function chosen was the time of flight, according to the mission requirement of rapid parcel transfer. Thus a time optimal problem was solved aiming to attain the optimal initial flight path angle, γ_0 . The latter is the main driver since the steering is only performed through gravity turn which is very sensible to the initial flight path angle γ_0 . While the constraint were set to be:

- dynamic pressure $q < 90$ kPa
- thermal load $q_t < 1 \frac{kW}{cm^2}$

5.3.1 Trajectory Optimization Results

Having the two approaches different free variable the results are slightly different, In the *brute force* approach we pursued the maximum possible range achieving:

Range	9789	km
γ_0	83.95	°
h_{turn}	12	m

Table 5.2: Optimal Trajectory Output (*Brute Force*)

Even if this range is theoretically achievable such a low h_{turn} is unfeasible since it would occur too close to the launch pad.

The outputs of the *Lagrangian Optimization* was instead:

Downrange	Range 1	Range 2	Range 3	Unit
	5498	7899	9094	[km]
γ_0	■	■	■	[°]
t_{flight}	2765	2612	2490	[s]

Table 5.3: Optimal Trajectory Output (*Lagrangian*)

More Plot showing that the dynamic pressure 8.7 and thermal load 8.8 constraints are satisfied

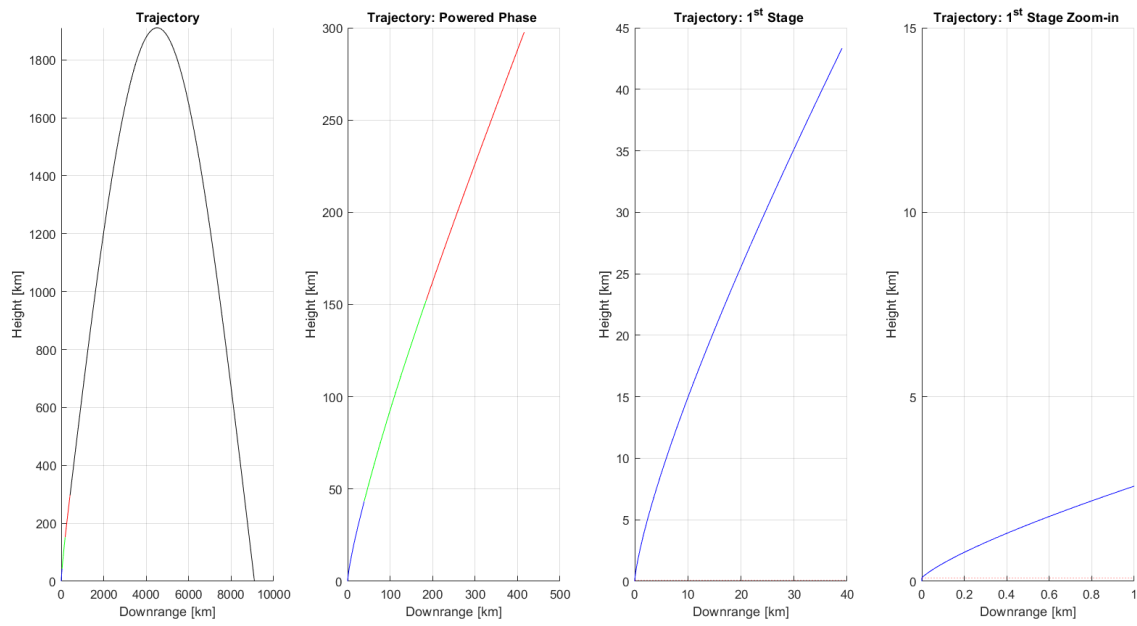


Figure 5.2: MAX Range Trajectory

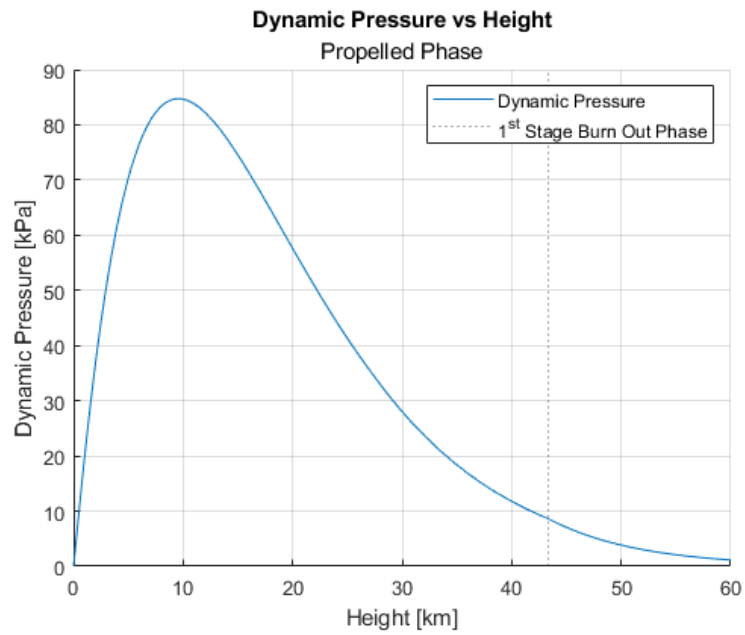


Figure 5.3: Dynamic Pressure in MAX Range Trajectory

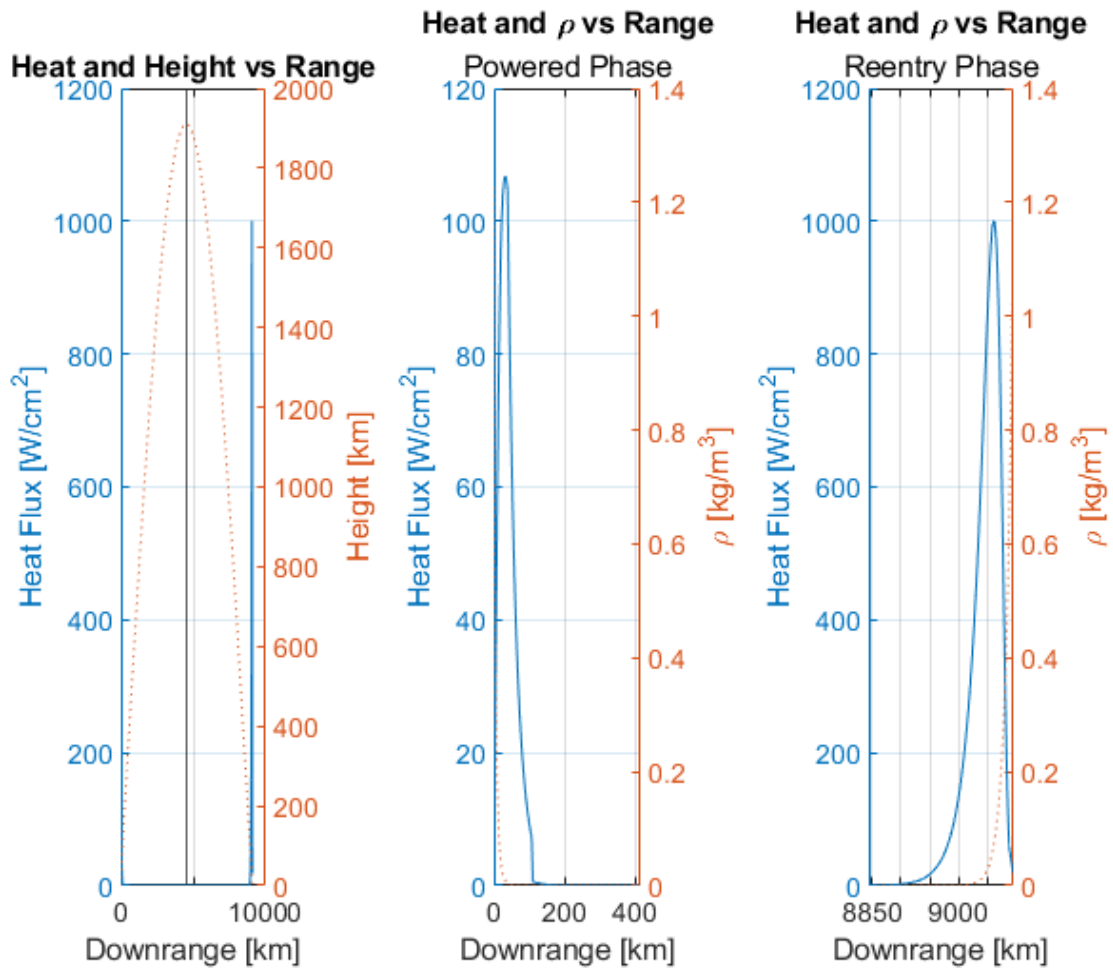


Figure 5.4: Thermal Load Plot in MAX Range Trajectory

5.4 Control

In the trajectory computation, the launcher was assumed as a point mass, but in reality, the full rocket's attitude must change to follow the trajectory and reach its destination. To accomplish this, a thrust vector control scheme was used. The thruster applies steering by deviating the thrust in the side direction of a steering angle δ , delimited by the propulsion unit in use, to modify the launch vehicle's state. In the case of Orion 50XL, δ is limited to $\pm 5^\circ$, being a solid booster the amount of control steering is lower than in liquid rockets, so the control action may take more time to correct the instabilities. The control analysis works as follows: as the rocket starts, the flight path angle is high, but as it follows the trajectory path, the flight path angle reduces. The velocity is tangent to the trajectory, so the rocket must change its orientation to have velocity alignment with the thrust at each time instant. This means that an angle of attack is present at each time and must be reduced to have thrust aligned with the velocity and minimize drag. The rocket is modeled as stated: This scheme shows that the forces get aligned with the body axis,

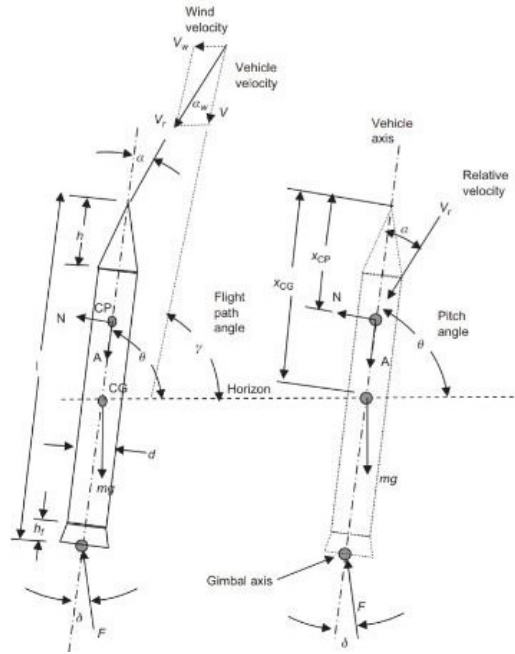


Figure 5.5: Workflow Overview

but change their application center: aerodynamic forces were positioned at the center of pressure, thrust was placed at the base, and weight at the center of mass. This system was described by the subsequent state, where δ acted as a control variable:

$$\begin{aligned}
 \dot{\theta} &= \theta \\
 \ddot{\theta} &= M_{\alpha}\alpha + M_{\delta}\delta \\
 \dot{\alpha} &= -\frac{F-A}{mv}\theta + \dot{\theta} - \frac{N_{\alpha}}{m}\alpha + \frac{T}{mv}\delta \\
 \dot{z} &= z \\
 \ddot{z} &= -\frac{F-A}{mv}\theta - \frac{N_{\alpha}}{m}\alpha + \frac{T}{m}\delta
 \end{aligned} \tag{5.6}$$

These equations assumed δ and θ small to apply some simplifications on the equations, but because θ was not small, the analysis set as 0 the pitch at next time step, which was the target value. Moreover, the forces and masses were kept constant to not over

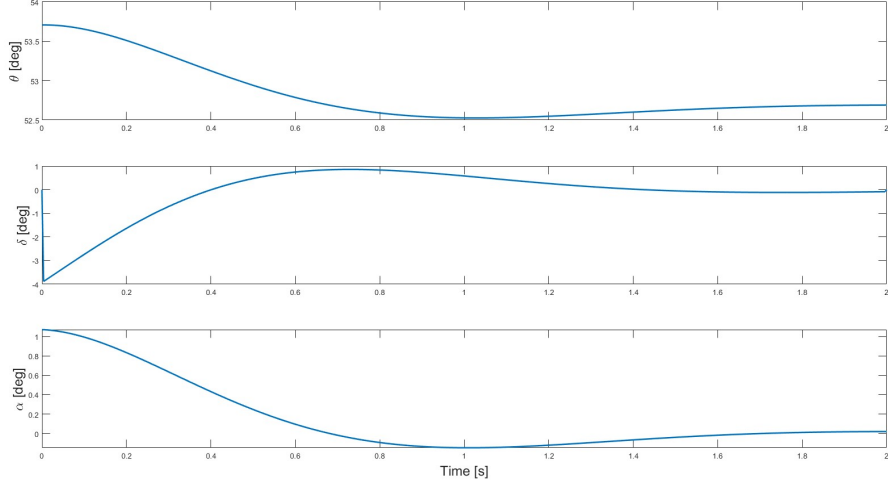


Figure 5.6: Workflow Overview

complicate the analysis unnecessarily. To maintain valid these assumptions, the control lasted for a small time step of 2 seconds, which was enough to reach the desired attitude. The analysis was carried out by solving an ordinary differential equation to reduce α to 0 so that θ becomes equal to γ . At each time step, a load minimum control law was used to determine the steering angle, where the coefficients used come from a best practice procedure:

$$\delta = -K2\dot{\theta} - K3\alpha \quad (5.7)$$

The steering angle grows over time to compensate the angle of attack and the angular velocity, meaning that as both angles grow, the momentum caused by the thrust raises proportionally, up until the point it starts to balance the vehicle and reduces α and $\dot{\theta}$ and so the steering attenuates. The control validation was performed on the problematic flight condition of maxq flight, so that if it works in the hard-to-control condition, it guarantees its functionality at less severe phases. The control can be graphically evaluated, where the highest value of the control obtained is -4° , meaning it stays below the threshold set by the propulsion unit of -5° and so the control law is validated.

Chapter 6

Structural Analysis

The structural analysis has been conducted following a straightforward design process since all the motors were selected from the market and did not require further modifications to their internal items to comply with the mission requirements. Therefore, the process is based on the design of the outer shell of the rocket, considering cylindrical shapes capable of withstanding all the different types of loads investigated for the purpose of this assignment. An initial analysis of the shear force and the bending moment which are acting along the launcher during its most critical conditions can be performed. It is decided to study two specific critical phases for the structural design. The first one occurs when the launcher is still on the launchpad, while the second one takes place during the flight, exactly when the launcher is subjected to the maximum dynamic pressure.

6.1 Ground loads

When the launch vehicle is prepared to be launched, the ground loads taken into account are the weight of the vehicle itself and the drag force produced by the wind at the launch base. For this computation, the rocket has been modeled as a beam clumped to the launch pad. The horizontal aerodynamic forces are applied as a distributed load on the entire lateral surface analyzed, while the axial load comes from the weight of the different items, listed in table [3.4].

The considered wind speed must also account for vortexes, for this reason, the assumed velocity is $v_w = 10m/s$ for a zero altitude air of $\rho = 1,225kg/m^3$. For simplicity, the nose is modeled with a cone shape, while the rest of the launch vehicle is modeled as a cylinder, with associated $C_d = 0.7$. Subsequently, the effective dynamic pressure q_w of the ground wind is calculated as:

$$q_w = \frac{1}{2}C_d\rho(v_w)^2S(x) \quad (6.1)$$

With this, the analysis proceeds to compute each axial, shear and bending load, starting from the nose and ending on the first stage, similarly to the scaling initialization. The results are shown below, where it is seen that the base is the location of the highest loads, in line with the assumptions of a clamped beam at the base model.

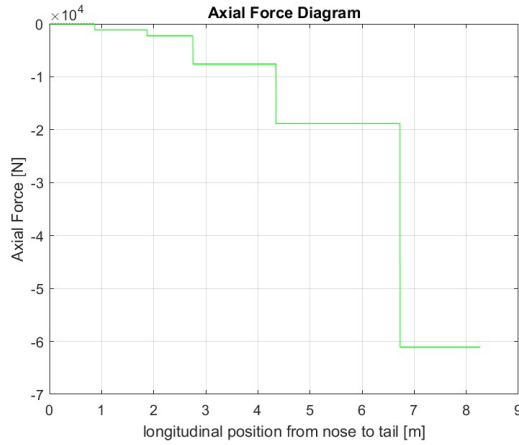


Figure 6.1: Axial loads

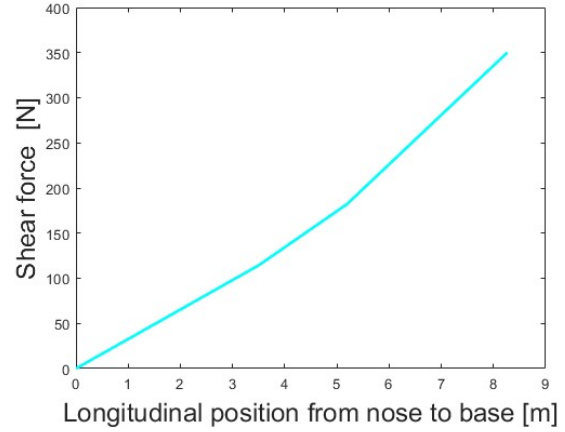


Figure 6.2: Shear loads

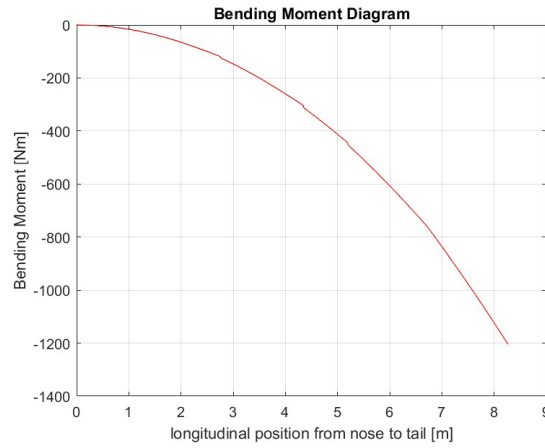


Figure 6.3: Bending loads

6.2 Flight loads

The launcher experiences maximum dynamic pressure during the first stack. In this condition, the shear force and bending moment in the structure can increase significantly. This can cause stress and deformation, which can lead to failure if the stress exceeds the yield strength of the chosen material, which has to be strong enough. Therefore, the effects of dynamic pressure is important during the design of structures that are subjected to supersonic or hypersonic fluid flow like in this case.

The model used for the structural analysis is the free-free beam. To obtain the shear force and the bending moment, the first step is to properly account for each individual structural component of the launcher: weight of the inert masses, aerodynamic forces and the thrust. The payload mass was assumed to be a concentrated force within the ogive, the remaining structure is divided in other four parts: the inert mass between the cone and the third stage and the three stages. Each of these masses was modeled as a localized load applied in the center of mass of each part in which the launcher was divided for this analysis. The aerodynamic forces were modeled as concentrated loads: one component of the lifting force applied in the center of pressure of the launcher, and the component of lifting force given by the fins applied in the position of the fins. The thrust was modeled

as a localized force acting on the base of the launcher. Every forces is decomposed in two components, one along the normal and one along the axial direction of the launcher. The shear forces and the bending moments are obtained from an equilibrium including the components mentioned above and the inertia forces to which the launcher is subjected. The following balance are shown:

$$m\ddot{\theta}(x_{cg,i} - x_{cg}) = -N - N_{fins} + T\cos(\delta) + m_i g \cos(\theta) \quad (6.2)$$

$$I\ddot{\theta} = N(x_{cp} - x_{cg}) - N_{fins}(x_{fins} - x_{cg}) + T\sin(\delta) \quad (6.3)$$

6.3 Thickness estimation

The design objective is to determine the thickness of the structure. As it can be seen in the table [6.1] the maximum values for axial force and bending moment are obtained for the case of maximum dynamic pressure. Therefore, it was decided to model the thickness

	On ground	At max-q
Maximum Axial Force [N]	████████	████████
Maximum Bending Moment [Nm]	██████	██████

Table 6.1: Maximum axial force and moment

based on this result, which is the most critical one. The value of the thickness is computed following the approach suggested in [13] by considering the critical stress of the material (equations 6.4). Regarding the material, it has been selected the Titanium alloy 6Al-4V [6]. This choice comes from the motor used for the design, since the second and third stage motors have already a titanium case, the same ones are kept in this analysis and are simply validated, thus the aluminum alloy is discarded. For the first stage, to have compliance with the other two stages, titanium is kept as well to have a uniform material case throughout the whole structure.

In this structure, the goal is to achieve the lightest possible solution to support the loads derived from the previous analysis. This process is explained in detail in the following paragraph.

The thickness to stand bending and axial loads is computed as shown in eq. 6.5.

$$t_{bend} = \frac{m_{bend}}{\pi r^2 \sigma_{yield}} \quad \text{and} \quad t_{axial} = \frac{4r\sigma_{yield}}{E} \quad (6.4)$$

The aggregation of the thickness components has been done directly by employing a conservative approach 6.5:

$$t = t_{bend} + t_{axial} \quad (6.5)$$

From this procedure, a thickness value of ██████ was found out.

Chapter 7

Reentry Phase

This section provides a qualitative overview over the reentry phase and a proposal for the reentry procedure. Further and more detailed evaluation should be carried starting from the considerations here presented.

7.1 Reentry Procedure

The mission aims to deliver a parcel to a target ground area, thus a reentry phase in the atmosphere has to be considered. During the reentry phase, the impact with the atmosphere leads to high thermal loads and dynamic pressure hence proper procedure should be proposed.

Some Preliminary Concepts

- **Entry Interface:** As the missile reaches the upper atmosphere, it encounters extreme temperatures due to friction. The entry interface is the point at which the vehicle enters the Earth's atmosphere. At this stage, temperatures can reach several thousand degrees Celsius due to the compression of air in front of the vehicle;
- **Thermal Protection System:** The reentry vehicle is equipped with a robust thermal protection system (TPS) that shields it from the intense heat. This system prevents the vehicle's structure and payload from reaching damaging temperatures. The TPS typically consists of ablative materials that erode gradually, dissipating heat and protecting the vehicle;
- **Aerodynamic Stabilization:** The vehicle's shape and design facilitate aerodynamic stabilization. The heat shield also contributes to maintaining the vehicle's orientation, ensuring a stable descent through the atmosphere;

7.2 Thermal Shielding

A first coarse estimation of thermal load can be performed accounting for the heat shield curvature radius. Since the equation [5.5] is meant for manned capsule with their peculiar geometry, some assumption were done. First of all, in the upward phase the ogive radius defined below in the equation is used as heat shield curvature radius R_N

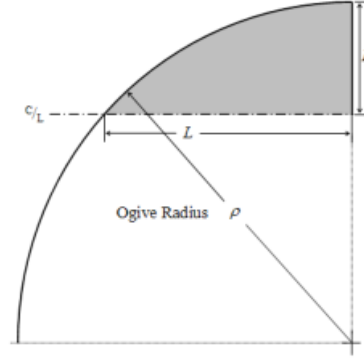


Figure 7.1: Cross Section of the Ogive

$$\rho = \frac{R^2 + l^2}{2R} = R_N \quad (7.1)$$

The reentry phase leads to the highest temperature, thus particular attention is used on this part. Since this choice for R_N is just an assumption and it does not account for the actual thermal stress concentration on the edge of the ogive, for the reentry phase a different and more conservative approach is chosen. The equation [5.5] considers the thermal load at a stagnation point on a curved heat shield thus for the highly demanding reentry phase a bottom-down reentry configuration is used. The reentry capsule is thus modeled as a ogive (pointing up) with a curved bottom [7.2]. The curvature radius is determined as:

$$R_N = 1 \text{ m} \quad (7.2)$$

This value, according to the expertise, can be considered the optimal choice accounting for both the best practice of having $1 \text{ m} \leq R_N \leq 4 \text{ m}$ [13] and the missile size and geometry.

Given these considerations two main materials were proposed for the thermal shielding of the missile: PICA and AVCOAT.

Here some peculiar characteristics of the materials:

- Thermal Conductivity (k): This property describes how well a material conducts heat. For ablative materials like PICA or AVCOAT, thermal conductivity can vary. It's often in the range of 0.1 to 0.5 $\frac{W}{mK}$ for such materials;
- Density ρ : The mass per unit volume of the material. Ablative materials usually have densities ranging from 100 to 1500 $\frac{kg}{m^3}$, depending on their composition;
- Specific Heat (C): It signifies the amount of heat required to raise the temperature of a unit mass of the material by 1 degree Celsius. For ablative materials, specific heat might range between 0.1 to 1 $\frac{kJ}{kgK}$;
- Ablation Properties: These properties include the material's ablation rate, which determines the rate at which the material erodes or dissipates during reentry due to heat. Ablation rates can vary widely based on the specific material and the reentry conditions.

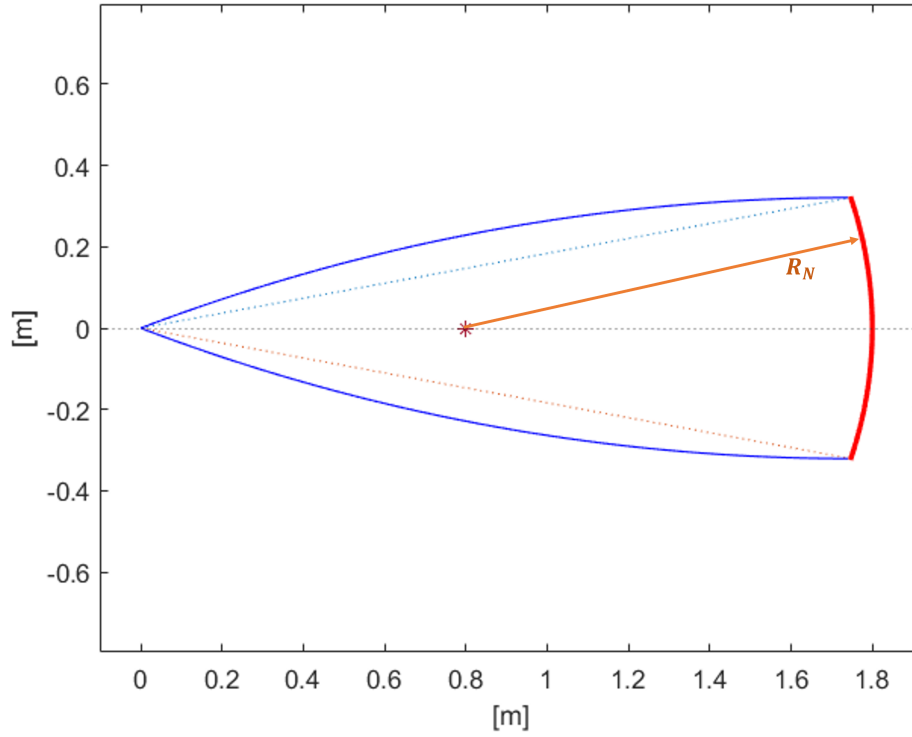


Figure 7.2: Cross Shape of the Capsule

7.3 Drogue and Parachute Deployment

The drogue parachute plays a critical role in the reentry process by stabilizing and decelerating the reentry vehicle before the main parachute deployment. To estimate the drogue parachute's geometry:

1. The terminal velocity of the reentry vehicle at the point of drogue parachute deployment must be considered. Typically, during the initial phase of reentry, the vehicle experiences extremely high speeds, often reaching several kilometers per second. As it enters the thicker part of the Earth's atmosphere, air resistance begins to slow it down;
2. Calculate the drag force required to decelerate the vehicle to a safe speed for main parachute deployment using the drag force equation:

$$F_{drag} = \frac{1}{2} (C_{d_{drogue}} A_{drogue} + C_{d_{capsule}} A_{capsule}) \rho v^2 \quad (7.3)$$

This equation represents the total drag force experienced by an object moving through air based on different drag coefficients and areas associated with specific sections of the objects, the drogue and the capsule itself, respectively;

3. Determine the drogue parachute's diameter and shape to generate sufficient drag force while ensuring stability during descent;
 - *Size and Shape:* The size and shape of the drogue parachute are crucial. A larger diameter parachute will generally generate more drag, slowing down

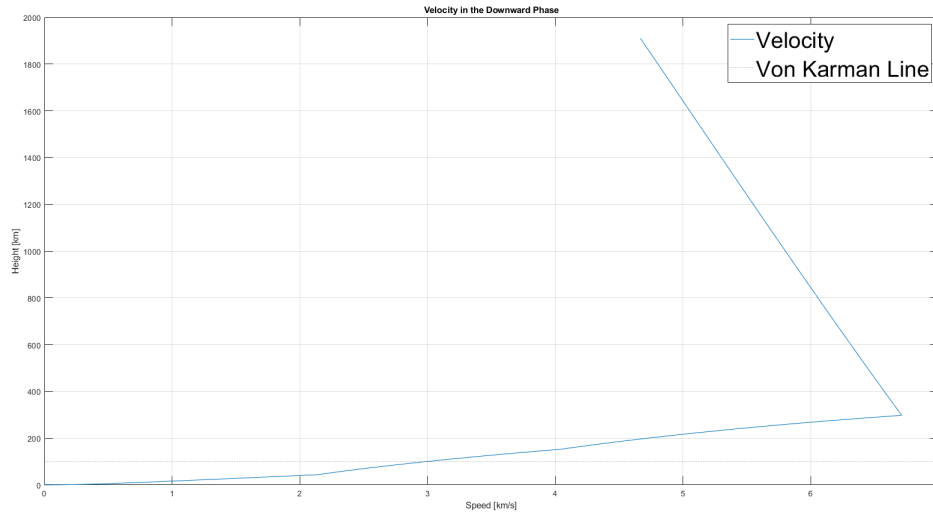


Figure 7.3: Height vs Velocity in the Downward Trajectory No Deceleration System

the descent. However, it is important to balance this with stability concerns. Drogue parachutes are often designed with a conical or disk shape to provide stability by minimising oscillations or spinning during descent;

- *Material and Construction:* The choice of materials used in the parachute fabric is essential for durability and performance. High-strength, lightweight materials are typically used to withstand the forces experienced during deployment and descent. Kevlar is a material known for its exceptional strength and heat resistance. It is often used in parachute construction, especially for components that experience higher stress or heat, such as suspension lines or reinforcement patches. Here the main features of the material:
 - *High Thermal Stability:* Kevlar exhibits remarkable thermal stability, retaining its mechanical properties even at elevated temperatures. It can withstand temperatures up to around 450°F (232°C) without significant degradation;
 - *Low Thermal Conductivity:* It possesses relatively low thermal conductivity, which means it's not a good conductor of heat. This property allows it to insulate against heat transfer effectively, protecting the material and structures made from it;
 - *Heat Resistance:* Kevlar's heat resistance is due to its aromatic polyamide structure. This structure provides excellent resistance to thermal degradation and maintains its strength even when exposed to high temperatures for extended periods;
 - *Fire Resistance:* While Kevlar is not inherently flame-resistant, it has a high resistance to burning and does not readily ignite. It chars rather than melts or drips when exposed to flames, providing some level of fire protection;
 - *Thermal Expansion:* Kevlar exhibits minimal thermal expansion under temperature variations. This characteristic is crucial in applications where dimensional stability under varying temperatures is essential;

- *Thermal Insulation*: In addition to its mechanical strength, Kevlar can serve as an effective thermal insulator, providing protection against heat transfer in applications where thermal insulation is necessary.

For these reasons Kevlar is selected for the future development of the drogue and parachute.

4. *Reefing System*: Some drogue parachutes are equipped with a reefing system that initially reduces the size of the parachute upon deployment and gradually allows it to fully inflate. This can help manage the initial shock of deployment and prevent excessive forces on the reentry vehicle. Integrating a reefing system into the reentry system is a strategic measure to optimize the deployment of the parachute, ensuring that the deceleration process aligns with the specific constraints on dynamic pressure and acceleration, thereby enhancing the overall safety and effectiveness of the reentry;
5. *Deployment Mechanism*: The deployment mechanism must ensure a controlled and reliable opening of the drogue parachute at the desired altitude and speed. This mechanism is critical to the parachute's functionality.

Given the payload mass and the designed system reentry vehicle dimensions, aerodynamic analysis and simulations can assist in estimating the appropriate drogue parachute size, shape, and deployment sequence. The deployment procedure should be carried out in two phases:

1. Drogue Parachute Deployment

- *Purpose*: The drogue parachute deployment occurs at a specific altitude and velocity to stabilize the vehicle and slow down its descent, preparing for the main parachute deployment;
- *Altitude and Velocity Criteria*: The drogue chute is usually deployed at higher speeds and altitudes to control the vehicle's descent before the main chute deployment. The exact parameters depend on the design specifications and mission requirements;
- *Stability Control*: The drogue parachute stabilizes the vehicle by reducing spin and ensuring a controlled descent. It also helps in proper positioning for the main parachute deployment.

2. Main Parachute Deployment

- *Timing*: Before the drogue chute stabilizes the descent, using the equation $t = \frac{V_{initial}}{g}$ This gives an approximate time it takes for the vehicle to reduce its velocity due to a deceleration of 1 g;
- *Deceleration*: Considering the deceleration due to the drag force exerted by the drogue chute. The drogue chute's drag force F_{drag} can be used in the simple equation

$$F_{drag} = ma \longrightarrow a = \frac{F_{drag}}{m} \quad (7.4)$$

And then using the equation [7.5] to compute the final velocity in order to

estimate the time it takes for the drogue chute to stabilize the decent with further computation.

$$v_{final}^2 = v_{initial}^2 + 2ad \longrightarrow a = \frac{F_{drag}}{m} \quad (7.5)$$

Where d is the distance covered under constant acceleration from the initial to the final velocity.

- *Timing:* After the drogue chute stabilizes the descent, the main parachute expansion occurs at a predetermined altitude and velocity lower than the drogue chute deployment criteria. The time it takes for the vehicle to reach a predetermined altitude from a stabilized descent altitude can be calculated using the following equation of motion [7.6]:

$$v^2 = u^2 + 2as \quad (7.6)$$

Where:

- v = final velocity at the predetermined altitude
- u = initial velocity (which could be zero if the vehicle is momentarily stationary at the stabilized descent altitude)
- s = Altitude difference between the stabilized descent altitude and the predetermined altitude

Solving for time:

$$t = \frac{v - u}{a} \quad (7.7)$$

7.3.1 Parachute Deployment Methods

The drogue parachute can be released through a multi-stage process. Upon reaching the designated altitude, the vehicle's deployment system initiates the drogue parachute's release. This parachute will rapidly eject and stabilize the vehicle's descent, enabling controlled navigation through the atmosphere. The main deployment systems are:

- *Pyrotechnic Release:* The ejection sequence often involves pyrotechnic devices or explosive charges that initiate the release of the parachutes. These devices ensure rapid and reliable ejection of the chutes at the precise moments required during descent;
- *Sequenced Activation:* The deployment system is designed to follow a precise sequence. The drogue chute ejects first to stabilize the vehicle, followed by the main parachute release once stability is achieved;
- *Redundancy and Reliability:* Redundancy is built into the deployment systems to ensure that if one chute fails to expand, a backup system can activate, preventing catastrophic failures and ensuring the safe return of the payload.

The success of the reentry and parachute unfolding procedures depends on extensive testing, simulations, and meticulous engineering to ensure reliable and safe operations under various conditions and contingencies. These procedures are crucial to ensure that the missile's payload returns safely to its intended destination. Further studies are required to evaluate the best choice for the deployment system.

7.3.2 Payload Recovery

A target zone should be chosen in advance. A tracking system shall be provided in order to keep trace of the payload and compute with a gradually increasing confidence the actually landing area, accounting for any perturbation occurring during the flight. A dedicated recovery team shall be on standby to swiftly retrieve the vehicle upon landing, ensuring the timely and secure retrieval of the payload.

7.4 Reentry Control

In order to prevent any toxicity-related problem, the mission is designed to produce all the required Δv in the very beginning of the flight, with a burn out time of $t_{\text{burn,tot}} =$,thus to avoid any issue in the descending phases onto the inhabited area. This mission requirement excluded the possibility of any propelled device for the control, pushing toward an aerodynamic control system. Given the necessity of an accurate maneuverability to keep the proper reentry configuration and compensate for random disturbances taking place during the flight. In the subsequent chapter regarding the future development of the project, a preliminary discussion of the grid fins is carried, out since they were recognized as a suitable and possibly well-performing solution for this design. Grid Fins could be use in the high atmosphere up the lower altitude before the deployment of the parachute, to exploit their optimal performances at hypersonic and supersonic speed. For lower speed and altitude, steerable parachutes could have been considered, but they were discarded because of their low TRL and limited testing.

Chapter 8

Final Considerations

8.1 Future Development

8.1.1 Grid Fins

Grid fins are a quite promising technology with a TRL 8 to 9. They are particularly suitable for hyper-sonic flight and their functionality for the control of the reentry. This technology, even if it is still under study, promises to be very reliable and performing as shown by the Space-X's Falcon 9, which uses grid fins to control the reentry achieving a perfect vertical landing. Other examples of application are the Soyuz Capsule and many missiles as: Vympel NPO R-77 and BGU-43 MOAB. Grid fins are also known



Figure 8.1: Grid Fins on Falcon 9

as *lattice fins* since their structure is essentially an assembly of small wings with a little chord. It was experimentally proven that those structures well behave in both hypersonic, supersonic and subsonic regimes while they have poor performances during the transonic phase. More specifically:

- *Hypersonic flow*: air goes through the lattice at high speed and isn't annoying at all because the shockwaves meet behind the fin;
- *Transonic air stream*: air is going to bounce on the grids and generate a lot of drag and force on the fin;
- *Close to Mach 1*: Grid Fins aren't really useful, because the shockwave is located before the grid fins, and no air stream through it equals stalling;

- *Subsonic speeds*: brings the air back in the grid fins, and give control of the rocket again. Aerodynamic moments can now orient the rocket to land it during her final approach,

We suggest further aerodynamic studies to exploit the lattice fins in the hypersonic part of the atmospheric reentry, in order to control the capsule, compensate for trajectory perturbations and eventually to slightly increase the drag force before the deployment of the drogue/parachute system. Due to the high thermal loads experienced in the reentry we advise the use of titanium for their manufacturing as already done by Space-X.

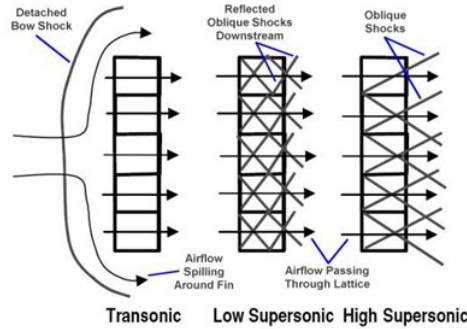


Figure 8.2: Behaviour of the Grid Fins at Different Velocity Regimes

8.2 Final Comment

According to our expertise, further studies should be performed on the aerodynamic of the launcher to investigate the feasibility of a grid fins control to stabilize, control and slow down the re-entering capsule. After an accurate evaluation of the reentry path quantitative evaluations shall be carried to size the drogue/parachute system and the thermal protection system. Moreover, a high frequency control shall be implemented to stabilize the second stack through TVC, while the third does not require any stabilization being its flight path out of the atmosphere.

Finally, a post-boost system must be implemented to tilt the capsule at the end of the powered phase, thus to prepare it to the final reentry configuration, avoiding to overfly inhabited areas during the downward phase, carrying fuel and thus minimizing the risk for the population.

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Authors Contributions

Chaari Ramy	<i>Panel method implementation and analysis. ICT support.</i>
Lugli Arianna	<i>Scaling and structural analysis and design</i>
Migadde Nicholao Mbalire	<i>Competitor analysis and reentry phase description</i>
Pratola Oscar	<i>Trajectory code implementation, trajectory analysis and optimization, reentry analysis. System engineering</i>
Rodriguez Alberto Lorenzo	<i>CFD aerodynamic analysis</i>
Spinelli Giulia Leda	<i>Requirement definition, market analysis, scaling and structural analysis and design. Project management</i>
Verri Matteo	<i>Control and scaling</i>

Appendix

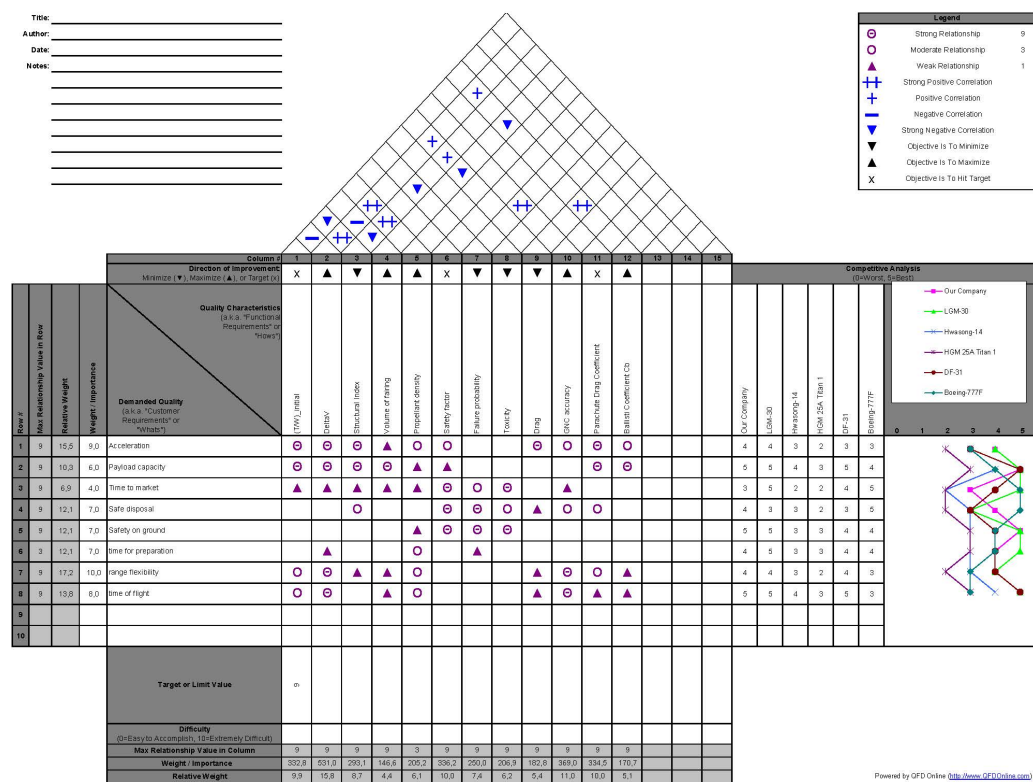


Figure 8.3: House of quality

Material [—]	σ_{yield} [GPa]	E [MPa]
Titanium	1030	113

Table 8.1: Titanium alloy properties [6]

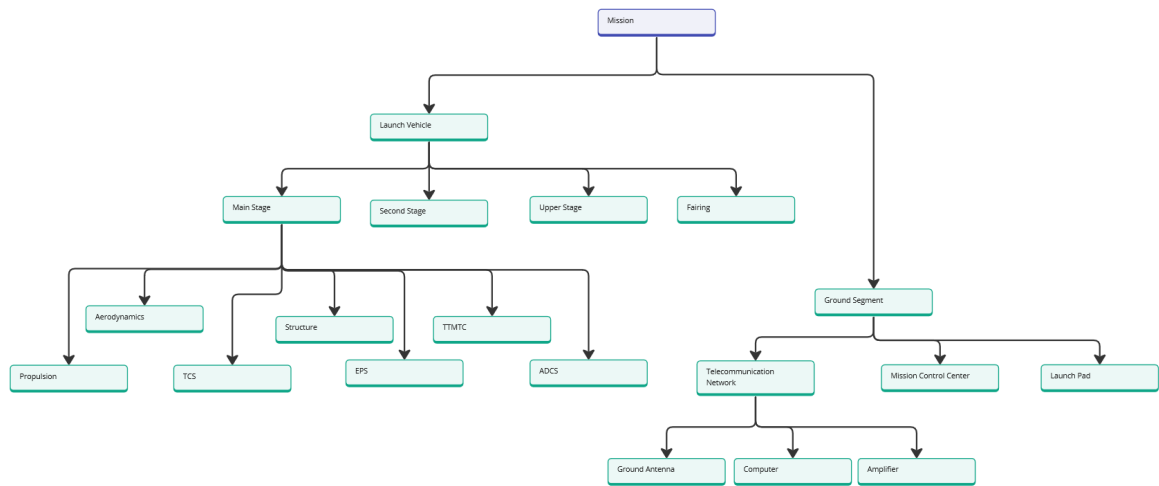


Figure 8.4: Product tree

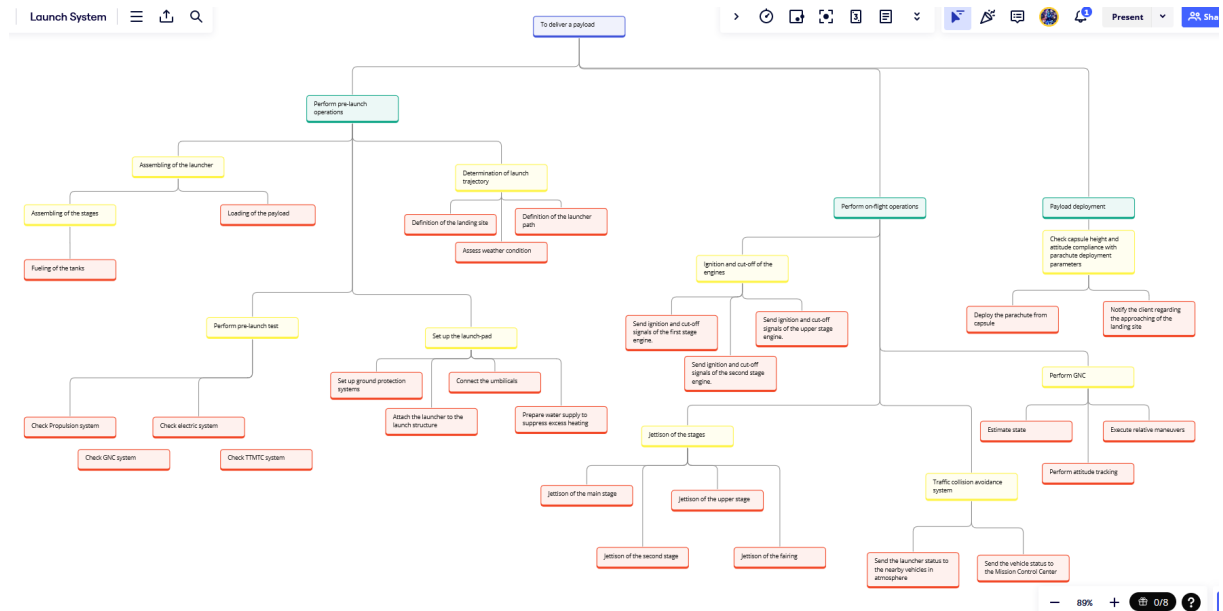


Figure 8.5: In-depth functional tree

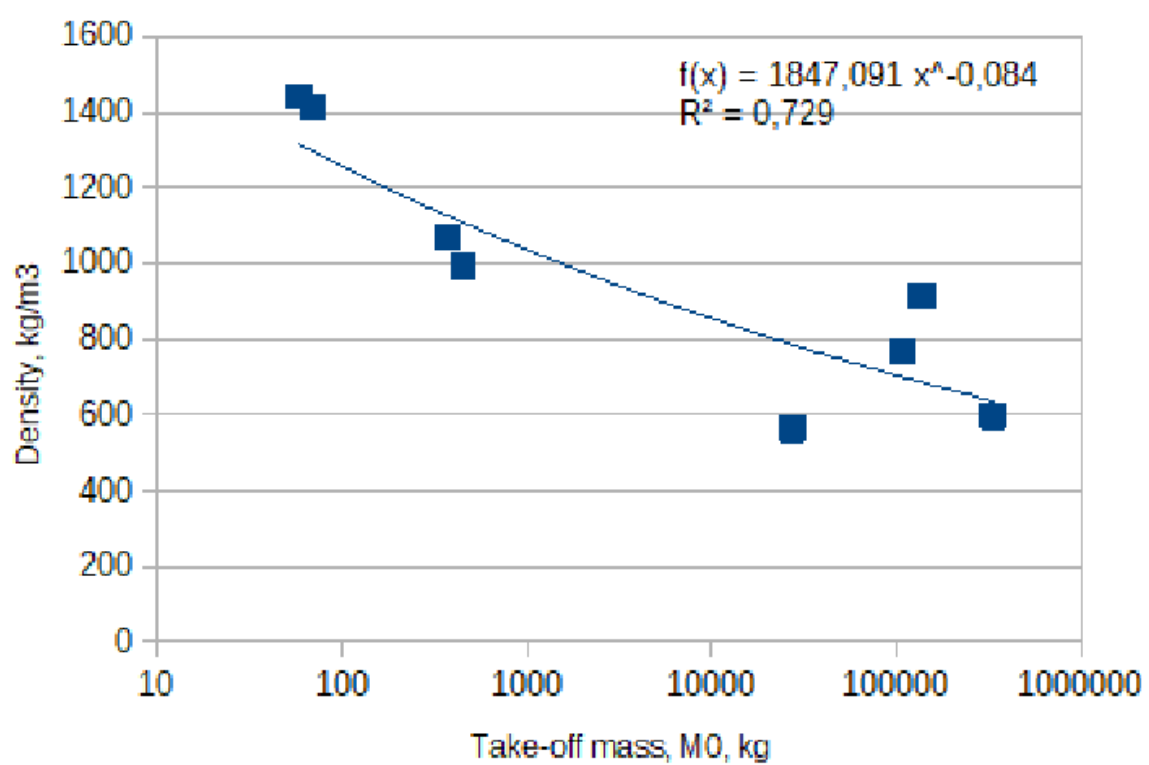


Figure 8.6: Missile density fitting law

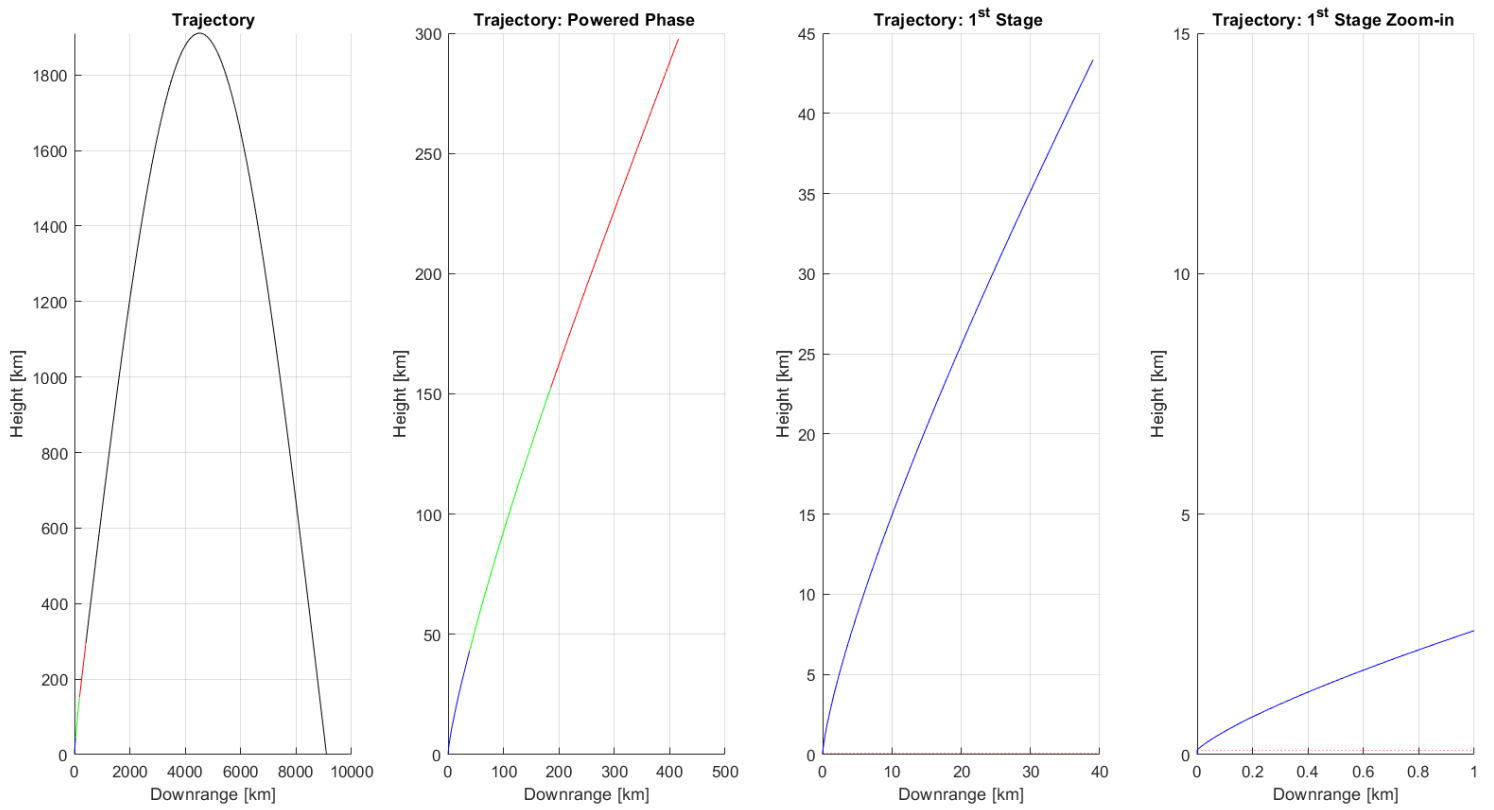


Figure 8.7: Trajectory

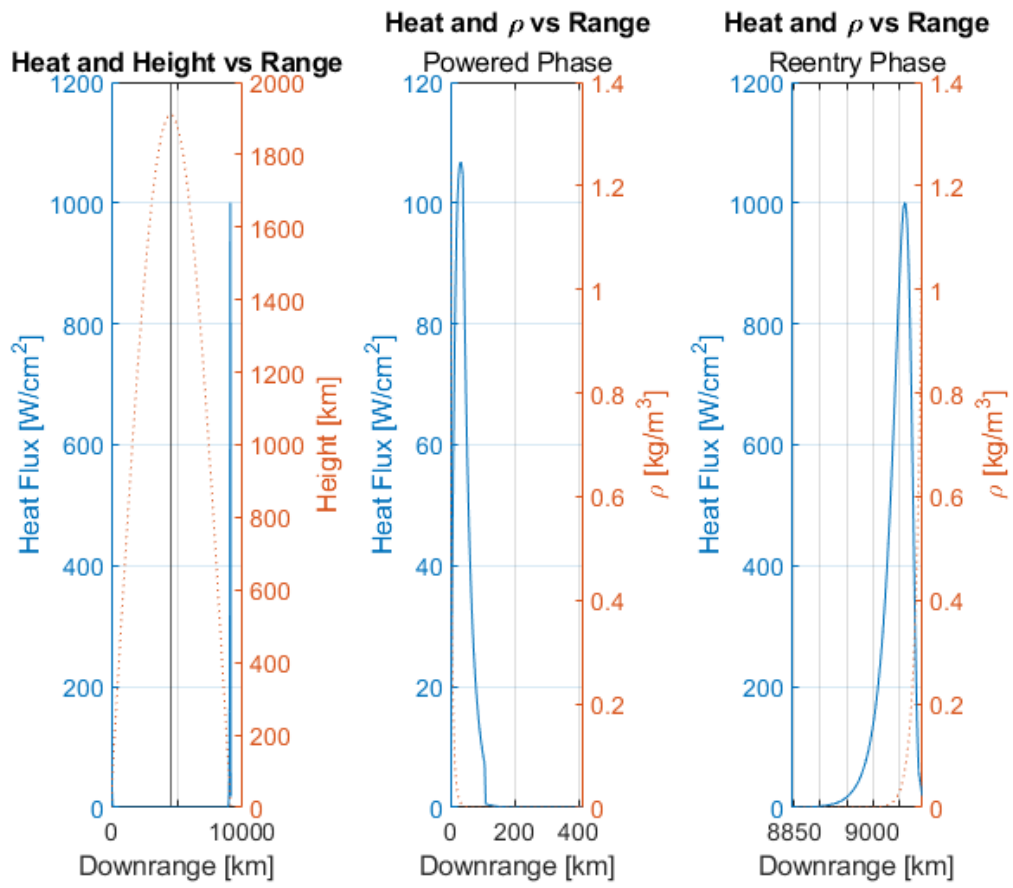


Figure 8.8: Thermal Load Plot